ALGEBRA II · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

84 question sets • 2525 questions

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Write complex numbers in the form a + bi

MAIN	Algebra II main question	**HSN-CN.A.1**
BACKWARD 1	Simplify square roots of positive numbers	**HSN-RN.A.2**
BACKWARD 2	No real number squares to a negative number	**8.EE.A.2**
BACKWARD 3	Divide each term of a sum by a number	**6.EE.A.3**
FORWARD 1	Use equality of complex numbers to find unknown real values	**HSN-CN.A.1**
FORWARD 2	Represent complex numbers as points in the complex plane	**HSN-CN.B.4**

SET 1 • HSN-CN.A.1

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-49)}$ in the form bi, where b is a real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q1/5

Write $\sqrt{(-18)}$ in the form bi with b in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q2/5

Write $6 - \sqrt{(-20)}$ in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q3/5

Write $(-4 + \sqrt{(-75)}) / 2$ in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q4/5

Write $(10 + \sqrt{(-64)}) / 4$ in the form a + bi. Then state its real part and its imaginary part.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 1

Simplify square roots of positive numbers • High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2

Simplify $\sqrt{50}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q1/5

Simplify $\sqrt{72}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q2/5

Which is √12 in simplest radical form?

A. 4√3

B. 3√2

C. 6√2

D. 2√3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q3/5

True or false?

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q4/5

Simplify $\sqrt{28}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

High School — Algebra I • Number & Quantity: The Real Number System • HSN-RN.A.2 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 2

No real number squares to a negative number • Grade 8 • Expressions & Equations • 8.EE.A.2

Find all real numbers x with $x^2 = 81$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q1/5

True or false?

$(-3)^2 = -9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q2/5

Is there a real number x with $x^2 = -4$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q3/5

Which number has a square equal to 36?

A. 18

B. $\sqrt{-36}$

C. -6

D. -18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q4/5

Evaluate $(-7)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSN-CN.A.1

BACKWARD 3

Divide each term of a sum by a number • Grade 6 • Expressions & Equations • 6.EE.A.3

Write $(8 + 6\sqrt{2}) / 2$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q1/5

Write $(-6 + 4\sqrt{5}) / 2$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q2/5

True or false?

$(10 + \sqrt{7}) / 5 = 2 + \sqrt{7}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q3/5

$(12 - 9\sqrt{3}) / 3$ equals

A. $4 - 3\sqrt{3}$

B. $4 - 9\sqrt{3}$

C. $4 - 3\sqrt{1}$

D. $12 - 3\sqrt{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q4/5

Write $(-4 + 2\sqrt{3}) / (-2)$ as a sum of two terms.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 1

Use equality of complex numbers to find unknown real values • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

x and y are real numbers and $(x - 3) + 2yi = 5 - 8i$. Find x and y.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q1/5

a and b are real numbers and $3a + (b + 1)i = -12 + 4i$. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q2/5

True or false?

If x and y are real and x + yi = 7, then y = 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q3/5

p and q are real and $(p + q) + (p - q)i = 10 + 2i$. What is p?

A. 6

B. 5

C. 4

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q4/5

m is a real number. For what value of m is $(m^2 - 4) + (m + 3)i$ a real number? What real number is it then?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSN-CN.A.1

FORWARD 2

Represent complex numbers as points in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4

In the complex plane the horizontal axis is real and the vertical axis is imaginary. Give the point for $-3 + 5i$.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q1/5

Which complex number is represented by the point $(4, -2)$ in the complex plane?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q2/5

Find the distance from 0 to the point 6 + 8i in the complex plane.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q3/5

True or false?

Every real number lies on the horizontal axis of the complex plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q4/5

Which number lies on the vertical axis of the complex plane?

A. -7

B. $7 + i$

C. $-7i$

D. $1 - i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S1-F2-Q5**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.4 • Set 1 • FORWARD 2 • Q5/5

Add and subtract complex numbers

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Combine like terms	**7.EE.A.1**
BACKWARD 2	Add and subtract integers and decimals	**7.NS.A.1**
BACKWARD 3	Write square roots of negative numbers with i	**HSN-CN.A.1**
FORWARD 1	Find real parameters that make a sum real or imaginary	**HSN-CN.A.2**
FORWARD 2	Add complex numbers as vectors in the complex plane	**HSN-CN.B.5**

SET 2 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Simplify (3 + 5i) + (−7 + 2i). Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q1/5

Simplify $(6 - 4i) - (-2 + 9i)$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q2/5

Simplify $(-1 + \sqrt{(-16)}) + (5 - \sqrt{(-9)})$. Write the answer in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q3/5

$z_1 = 2.5 - i$, $z_2 = -1.5 + 3i$ and $z_3 = 4i$. Find $z_1 - z_2 + z_3$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q4/5

Find the complex number w for which $(4 - 7i) + w = -2 + 3i$.

$w =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • MAIN • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 1

Combine like terms • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify $(3 + 5x) + (-7 + 2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q1/5

Simplify $(6 - 4x) - (-2 + 9x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q2/5

$(2 - 3y) - (5 - y)$ equals

A. $-3 - 2y$

B. $7 - 4y$

C. $-3 + 2y$

D. $-3 - 4y$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

$(4 + 2t) + (1 - 6t) = 5 + 4t$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q4/5

Simplify $(0.5 - x) - (1.5 + 2x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 2

Add and subtract integers and decimals • Grade 7 • The Number System • 7.NS.A.1

Evaluate $6 - (-2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q1/5

Evaluate $-4 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q2/5

$-1 - 3 + 4$ equals

A. 8

B. 0

C. -8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q3/5

True or false?

$2.5 - (-1.5) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q4/5

Evaluate $-2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 7 • The Number System • 7.NS.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSN-CN.A.2

BACKWARD 3

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-16)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q1/5

Write $\sqrt{(-9)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q2/5

$\sqrt{(-25)}$ equals

A. -5

B. $25i$

C. $-5i$

D. $5i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-B3-Q3**

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q3/5

True or false?

$\sqrt{(-1)} = i$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q4/5

Write $-\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 2 • BACKWARD 3 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 1

Find real parameters that make a sum real or imaginary • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

k is a real number. For what value of k is $(3 + ki) + (k - 5i)$ a real number? Give that real number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q1/5

k is real. For what value of k is
$(2k - 6) + 4i - (k + 1 - i)$ a pure imaginary number (real part 0)?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q2/5

True or false?

The sum of a complex number and its conjugate is always 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q3/5

a is real and $(a + 3i) - (2 + ai)$ is real. What is a?

A. 0

B. 2

C. 3

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q4/5

Explain why the sum of a + bi and a − bi is always a real number when a and b are real.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSN-CN.A.2

FORWARD 2

Add complex numbers as vectors in the complex plane • High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5

Plot 2 + i and 1 + 3i as points in the complex plane (real axis horizontal). What point represents their sum?

Sum point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q1/5

The points 0, 2 + i, 1 + 3i and their sum are vertices of a quadrilateral. What kind of quadrilateral is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q2/5

True or false?

Subtracting w from z gives the vector from the point w to the point z.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q3/5

What is the distance between the points $5 + 2i$ and $1 - i$ in the complex plane?

A. 25

B. $\sqrt{13}$

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S2-F2-Q4**

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q4/5

Find the midpoint of the segment joining $6 - 2i$ and $-2 + 4i$, as a complex number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Precalculus (course placement in this bank) • Number & Quantity: The Complex Number System • HSN-CN.B.5 • Set 2 • FORWARD 2 • Q5/5

Multiply complex numbers and simplify powers of i

MAIN	Algebra II main question	**HSN-CN.A.2**
BACKWARD 1	Multiply binomials	**HSA-APR.A.1**
BACKWARD 2	Use $i^2 = -1$ and the cycle of powers of i	**HSN-CN.A.1**
BACKWARD 3	Divide whole numbers to find a remainder	**4.NBT.B.6**
FORWARD 1	Find a real parameter that makes a product real	**HSN-CN.A.2**
FORWARD 2	Divide complex numbers using conjugates	**HSN-CN.A.3**

SET 3 • HSN-CN.A.2

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(3 + 2i)(4 - 5i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q1/5

Simplify $(-2 + i)^2$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q2/5

Multiply $(5 - 3i)(5 + 3i)$. Write the product in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q3/5

Simplify $(1 + i)^3$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q4/5

Simplify $i^{23} \cdot (4 - i)$. Write the answer in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 1

Multiply binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(3 + 2x)(4 - 5x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q1/5

Expand $(-2 + x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q2/5

$(5 - 3x)(5 + 3x)$ equals

A. $25 + 9x^2$

B. $10 - 9x^2$

C. $25 - 9x^2$

D. $25 - 30x - 9x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

$(1 + x)^2 = 1 + x^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q4/5

Multiply $x(4 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 2

Use $i^2 = -1$ and the cycle of powers of i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Simplify i^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q1/5

Simplify i^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q2/5

Simplify i^4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q3/5

True or false?

$-10i^2 = -10$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q4/5

Which equals 1?

A. i^{8}

B. i^{9}

C. i^{3}

D. i^{6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSN-CN.A.2

BACKWARD 3

Divide whole numbers to find a remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Find the remainder when 23 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q1/5

Find the remainder when 50 is divided by 4.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q2/5

What is the remainder when 36 is divided by 4?

A. 0

B. 4

C. 1

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q3/5

True or false?

The remainder when 101 is divided by 4 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q4/5

Write 39 as 4q + r with 0 ≤ r < 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 1

Find a real parameter that makes a product real • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

a is a real number. For what value of a is $(a + 2i)(3 - i)$ a real number? Give the product.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q1/5

b is real. For what value of b is (2 + bi)(1 + i) pure imaginary (real part 0)?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q2/5

True or false?

The product of two non-real complex numbers can be a real number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q3/5

For real c, $(c + i)^2$ is real when

A. $c = 0$

B. $c = 1$

C. $c = -1$

D. never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q4/5

Show that $(a + bi)(a - bi)$ is a real number for all real a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSN-CN.A.2

FORWARD 2

Divide complex numbers using conjugates • High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3

Write (3 + 4i) / (1 + 2i) in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q1/5

Write $10 / (3 - i)$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q2/5

True or false?

To divide by 2 − 5i, multiply numerator and denominator by 2 − 5i.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q3/5

(2 + i) / i equals

A. $-1 + 2i$

B. $2 + i$

C. $1 - 2i$

D. $1 + 2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q4/5

Explain why multiplying by the conjugate of the denominator always gives a real denominator.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Algebra II, Advanced Extension (+) • Number & Quantity: The Complex Number System • HSN-CN.A.3 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSN-CN.C.7 • THE COMPLEX NUMBER SYSTEM

Solve quadratic equations with complex solutions

MAIN	Algebra II main question	**HSN-CN.C.7**
BACKWARD 1	Use the quadratic formula and the discriminant	**HSA-REI.B.4.b**
BACKWARD 2	Write square roots of negative numbers with i	**HSN-CN.A.1**
BACKWARD 3	Solve equations of the form $(x - h)^2 = k$	**HSA-REI.B.4.b**
FORWARD 1	Find a quadratic from one complex solution	**HSN-CN.C.7**
FORWARD 2	Classify solutions for a family of quadratics	**HSN-CN.C.7**

SET 4 • HSN-CN.C.7

MAIN

Algebra II main question • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 49 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q1/5

Solve $x^2 - 4x + 13 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q2/5

Solve $2x^2 + 2x + 5 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q3/5

Solve $(x + 3)^2 = -20$ over the complex numbers. Give exact solutions in the form a + bi.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q4/5

Solve $3x^2 - 6x + 7 = 0$ over the complex numbers. Give exact solutions in the form a + bi.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • MAIN • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 1

Use the quadratic formula and the discriminant • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Find the discriminant $b^2 - 4ac$ of $x^2 - 4x + 13 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q1/5

Find the discriminant of $2x^2 + 2x + 5 = 0$.

Discriminant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q2/5

True or false?

If the discriminant is negative, the quadratic equation has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q3/5

In $3x^2 - 6x + 7 = 0$, the values of a, b and c are

A. 3, −6, −7

B. 3, 6, 7

C. −3, −6, 7

D. 3, −6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q4/5

Write the quadratic formula.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 2

Write square roots of negative numbers with i • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1

Write $\sqrt{(-36)}$ in the form bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q1/5

Write $\sqrt{(-48)}$ in simplest form with i.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q2/5

$\sqrt{(-20)}$ equals

A. $2\sqrt{5}\,i$

B. $4\sqrt{5}\,i$

C. $-2\sqrt{5}$

D. $10i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q3/5

True or false?

$\sqrt{(-49)} = -7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q4/5

Write $(-2 + \sqrt{(-36)}) / 4$ in the form a + bi.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSN-CN.C.7

BACKWARD 3

Solve equations of the form $(x - h)^2 = k$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $(x - 2)^2 = 9$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q1/5

Solve $(x + 1)^2 = 5$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q2/5

True or false?

Solving $x^2 = 16$ gives only $x = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q3/5

The solutions of $(x + 4)^2 = 25$ are

A. 5 and −5

B. −1 and 9

C. 1 and −9

D. 21 and −29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q4/5

Solve $x^2 = 50$ over the real numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 4 • BACKWARD 3 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 1

Find a quadratic from one complex solution • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

$3 - 2i$ is a solution of $x^2 + bx + c = 0$, where b and c are real. Find b and c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q1/5

$-1 + 4i$ is a solution of a quadratic equation $x^2 + bx + c = 0$ with real b and c. Write the equation.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q2/5

True or false?

If 2 + i is a solution of a quadratic with real coefficients, then 2 − i is also a solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q3/5

Which quadratic with real coefficients has 5i as a solution?

A. $x^2 + 25 = 0$

B. $x^2 + 5 = 0$

C. $x^2 - 25 = 0$

D. $x^2 - 10x + 25 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q4/5

Explain why $x^2 - (2 + i)x = 0$ has the solution $2 + i$ but not the solution $2 - i$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSN-CN.C.7

FORWARD 2

Classify solutions for a family of quadratics • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

For which real values of k does $x^2 + kx + 9 = 0$ have two non-real solutions?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q1/5

For which real values of c does $x^2 - 4x + c = 0$ have non-real solutions? Give them for c = 8.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q2/5

True or false?

$x^2 + kx - 1 = 0$ has non-real solutions for some real k.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q3/5

$x^2 + 6x + m = 0$ has exactly one (repeated) real solution when $m =$

A. 3

B. 36

C. -9

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q4/5

For $c > 4$, write the solutions of $x^2 - 4x + c = 0$ in terms of c.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 4 • FORWARD 2 • Q5/5

Factor polynomials over the complex numbers

MAIN	Algebra II main question	**HSN-CN.C.8**
BACKWARD 1	Factor a difference of squares	**HSA-SSE.A.2**
BACKWARD 2	Multiply complex conjugates	**HSN-CN.A.2**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Factor cubic and quartic polynomials over the complex numbers	**HSN-CN.C.8**
FORWARD 2	Write polynomials with real coefficients from complex zeros	**HSN-CN.C.9**

SET 5 • HSN-CN.C.8

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^2 + 25$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q1/5

Factor $x^2 + 7$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q2/5

Factor $4x^2 + 9$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q3/5

Factor $x^4 - 16$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q4/5

Factor $x^2 - 6x + 13$ into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • MAIN • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 1

Factor a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 49$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q1/5

Factor $9x^2 - 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q2/5

$x^4 - 1$ equals

A. $(x - 1)^4$

B. $(x^2 - 1)^2$

C. $(x^2 - 1)(x^2 + 1)$

D. $(x^2 + 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q3/5

True or false?

$x^2 + 9 = (x + 3)(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q4/5

Factor $x^2 - 5$ using radicals.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 2

Multiply complex conjugates • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Multiply $(x + 5i)(x - 5i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q1/5

Multiply $(2x + 3i)(2x - 3i)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q2/5

True or false?

$(3i)^2 = 9$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q3/5

$(x - (1 + i))(x - (1 - i))$ equals

A. $x^2 + 2x + 2$

B. $x^2 - 2x + 2$

C. $x^2 + 2$

D. $x^2 - 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q4/5

Simplify $(i\sqrt{7})^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSN-CN.C.8

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 25 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q1/5

Solve $x^2 - 6x + 13 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q2/5

True or false?

If r is a zero of p(x), then x − r is a factor of p(x).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q3/5

The zeros of $x^2 + 4$ are

A. $2i$ and $-2i$

B. $4i$ and $-4i$

C. -4

D. 2 and -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q4/5

Solve $x^2 + 2x + 5 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 1

Factor cubic and quartic polynomials over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^3 + 8$ completely into linear factors with complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q1/5

Factor $x^3 + 9x$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q2/5

Factor $x^4 + 5x^2 + 4$ completely into linear factors.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q3/5

True or false?

$x^3 - 1 = (x - 1)(x^2 + x + 1)$, and $x^2 + x + 1$ has two non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q4/5

Which is a linear factor of $x^3 - 27$ over the complex numbers?

A. $x + 3i$

B. $x + 3$

C. $x - 3i$

D. $x - (-3/2 + (3\sqrt{3}/2)i)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F1-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSN-CN.C.8

FORWARD 2

Write polynomials with real coefficients from complex zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Write a polynomial with real coefficients, leading coefficient 1 and least degree, whose zeros include $2i$ and $1 - i$.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q1/5

What is the least degree of a polynomial with real coefficients that has zeros 3, 4i and 2 + i?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q2/5

True or false?

A polynomial with real coefficients can have exactly one non-real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q3/5

A cubic with real coefficients has zeros 5 and $-i$. Its third zero is

A. -5

B. $5i$

C. 1

D. i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S5-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q4/5

Explain why a polynomial with real coefficients and degree 3 must have at least one real zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 5 • FORWARD 2 • Q5/5

SET 6 • HSN-CN.C.9 • THE COMPLEX NUMBER SYSTEM

Use the Fundamental Theorem of Algebra

MAIN	Algebra II main question	**HSN-CN.C.9**
BACKWARD 1	Find the degree of a polynomial	**HSA-SSE.A.1.a**
BACKWARD 2	Find zeros from factored form	**HSA-SSE.B.3.a**
BACKWARD 3	Solve quadratics with complex solutions	**HSN-CN.C.7**
FORWARD 1	Relate the complex zeros to the real x-intercepts of a graph	**HSA-APR.B.3**
FORWARD 2	Prove existence statements about real zeros	**HSN-CN.C.9**

SET 6 • HSN-CN.C.9

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

How many complex zeros, counted with multiplicity, does $p(x) = 3x^5 - x^2 + 7$ have? Name the theorem you used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q1/5

List every complex zero of $p(x) = (x - 2)^3(x^2 + 9)$ with its multiplicity, and check that the multiplicities add to the degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q2/5

Which argument shows that every quadratic $p(x) = ax^2 + bx + c$ (real a, b, c, $a \neq 0$) has exactly two complex zeros, counted with multiplicity?

A. $x^2 - 5x + 6 = (x - 2)(x - 3)$ has two zeros, so every quadratic has two zeros.

B. By the quadratic formula the zeros are $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. If $b^2 - 4ac < 0$ the square root does not exist, so there are no zeros; otherwise there are two, so a quadratic never has more than two zeros.

C. The graph of every quadratic is a parabola, and a parabola that opens up or down always crosses the x-axis in exactly two places, so every quadratic $ax^2 + bx + c$ has exactly two zeros.

D. The quadratic formula $(-b \pm \sqrt{(b^2 - 4ac)})/(2a)$ gives every zero: two real zeros if $b^2 - 4ac > 0$, one zero of multiplicity 2 if $b^2 - 4ac = 0$, and two non-real conjugate zeros if $b^2 - 4ac < 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q3**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q3/5

Find every complex zero of $p(x) = x^4 - 81$ and check that the count agrees with the Fundamental Theorem of Algebra.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q4/5

A student says $p(x) = x^3 + x + 1$ has no zeros because it does not factor with integers. Which statement is correct?

A. The student is right: a polynomial that does not factor over the integers has no zeros.

B. It has exactly 3 real zeros, because the Fundamental Theorem of Algebra says a polynomial of degree 3 has 3 zeros, and a polynomial with real coefficients has only real zeros.

C. It has 3 complex zeros counted with multiplicity, and at least one is real, because non-real zeros of a polynomial with real coefficients come in conjugate pairs.

D. It has 3 complex zeros counted with multiplicity, and all three could be non-real, because a polynomial that does not factor over the integers need not cross the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-M-Q5**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • MAIN • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 1

Find the degree of a polynomial • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

What is the degree of $3x^5 - x^2 + 7$?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q1/5

What is the degree of $(x - 2)^3(x^2 + 9)$ when multiplied out?

Degree: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q2/5

The degree of $x(x + 1)(x^2 - 4)$ is

A. 1

B. 4

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q3/5

True or false?

The degree of $4x^3 + 2x^4$ is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q4/5

What is the leading coefficient of $-2x^3 + 5x - 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 2

Find zeros from factored form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a

Find the zeros of $(x - 2)(x + 5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q1/5

Find the zeros of $(x - 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q2/5

True or false?

A product is 0 only when one of its factors is 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q3/5

The zeros of $x(x + 4)$ are

A. 0 and 4

B. 0 and -4

C. -4 only

D. 4 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q4/5

Find the zeros of $(2x - 3)(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.B.3.a • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSN-CN.C.9

BACKWARD 3

Solve quadratics with complex solutions • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^2 + 9 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q1/5

Solve $x^2 + 2x + 2 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q2/5

True or false?

$x^2 + 1 = 0$ has no complex solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q3/5

The solutions of $x^2 - 2x + 5 = 0$ are

A. $1 \pm 4i$

B. $-1 \pm 2i$

C. $1 \pm 2i$

D. $2 \pm i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q4/5

Solve $x^2 = -81$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 1

Relate the complex zeros to the real x-intercepts of a graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A polynomial of degree 4 with real coefficients has a graph that crosses the x-axis at exactly two points and touches it nowhere else. Each crossing is a simple zero. How many non-real zeros does it have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q1/5

The graph of $y = x^3 - 4x^2 + 5x - 2$ touches the x-axis at $x = 1$ and crosses it at $x = 2$. List all complex zeros with multiplicity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A polynomial of degree 6 whose graph never meets the x-axis has 6 non-real zeros.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q3/5

A degree-5 polynomial with real coefficients has exactly one real zero, of multiplicity 1. How many non-real zeros does it have?

A. 4

B. 5

C. 2

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q4/5

Can a degree-4 polynomial with real coefficients have exactly three real zeros, each of multiplicity 1? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSN-CN.C.9

FORWARD 2

Prove existence statements about real zeros • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9

Which argument proves that every polynomial of odd degree with real coefficients has at least one real zero?

A. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come one at a time, so there is an odd number of them, and an odd number of non-real zeros leaves at least one zero that is real.

B. By the Fundamental Theorem of Algebra it has n zeros counted with multiplicity, n odd. Non-real zeros come in conjugate pairs, so there is an even number of them; odd minus even is odd, so at least one zero is real.

C. The Fundamental Theorem of Algebra says that every polynomial of degree $n \geq 1$ with real coefficients has at least one real zero; a polynomial of odd degree has degree at least 1, so it has at least one real zero as well.

D. $x^3 - 1$ has the real zero 1, so every polynomial of odd degree has a real zero.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q1/5

A degree-4 polynomial with real coefficients has the zeros 1 + i and 3. What are all of its possible zero lists?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q2/5

True or false?

A degree-4 polynomial with real coefficients must have at least one real zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q3/5

A polynomial with real coefficients has degree 7. The possible numbers of real zeros (with multiplicity) are

A. 1, 3, 5, 7

B. 7 only

C. 0, 2, 4, 6

D. 0 to 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S6-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q4/5

Explain why the Fundamental Theorem of Algebra says nothing about whether a polynomial's zeros can be found exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.9 • Set 6 • FORWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a • SEEING STRUCTURE IN EXPRESSIONS

Interpret the parts of polynomial, rational, and exponential expressions in context

MAIN	Algebra II main question	**HSA-SSE.A.1.a**
BACKWARD 1	Name terms, factors, and coefficients	**HSA-SSE.A.1.a**
BACKWARD 2	Translate between a percent change and a multiplying factor	**7.RP.A.3**
BACKWARD 3	Interpret a quotient as an amount per unit	**6.RP.A.2**
FORWARD 1	Build an expression whose parts have given meanings	**HSA-SSE.A.1.a**
FORWARD 2	Rewrite an exponential expression to reveal a different rate	**HSF-IF.C.8.b**

SET 7 • HSA-SSE.A.1.a

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

A town's population is modeled by $P(t) = 12{,}500(0.97)^t$, where t is the number of years after 2020. What do 12,500 and 0.97 represent, and what does $1 - 0.97$ represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q1/5

The average cost per item of making x items is $C(x) = (4{,}000 + 6x) / x$ dollars. Interpret 4,000, 6x, and the division by x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q2/5

An open box is made by cutting squares of side x cm from the corners of a 20 cm by 20 cm sheet. Its volume is $V(x) = x(20 - 2x)^2$ cm³. Interpret x, $20 - 2x$, and $(20 - 2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q3/5

The amount of a medicine in the body t hours after a dose is $A(t) = 200(1/2)^{t/6}$ mg. Interpret 200, 1/2, and t/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q4/5

Maya keeps $1,500 in an account that earns 4% interest compounded once a year and also puts $200 per year in a jar (no interest). Her total after n years is $T(n) = 1500(1.04)^n + 200n$. Interpret $1500(1.04)^n$, 1.04, and 200n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • MAIN • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 1

Name terms, factors, and coefficients • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Name the terms of 4,000 + 6x.

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q1/5

What is the coefficient of x in $7x - 3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q2/5

In $5x(x - 2)$, which is a factor?

A. $5x - 2$

B. $x - 10$

C. $x - 2$

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q3/5

True or false?

In 3 + 2x, the number 3 is a factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q4/5

Name the factors of $200(0.5)^t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 2

Translate between a percent change and a multiplying factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A price is multiplied by 0.97. By what percent does it change?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q1/5

What factor multiplies an amount that increases by 4%?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q2/5

Multiplying by 1.25 is the same as

A. increasing by 2.5%

B. increasing by 25%

C. increasing by 125%

D. decreasing by 25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q3/5

True or false?

Multiplying by 0.8 decreases an amount by 80%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q4/5

Find $1 - 0.85$ and explain what it means for an amount multiplied by 0.85.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSA-SSE.A.1.a

BACKWARD 3

Interpret a quotient as an amount per unit • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A total cost of $600 is shared by 30 people. What does 600/30 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q1/5

A car travels 150 miles in 3 hours. What does 150/3 represent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q2/5

If T is a total cost for n items, then T/n is

A. the profit

B. the number of items

C. the cost per item

D. the total cost

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Dividing a total by the number of items gives an average per item.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q4/5

What does (4,000 + 6x)/x mean when x = 100?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 1

Build an expression whose parts have given meanings • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write an expression for the value of a $24,000 car that loses 15% of its value each year, t years after purchase.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q1/5

A gym charges a $90 joining fee and $35 per month. Write an expression for the average cost per month over m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q2/5

True or false?

An expression for a quantity that triples every 5 years, starting at 40, is $40(3)^{5t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q3/5

A bacteria count starts at 500 and doubles every 3 hours. Which expression gives the count after h hours?

A. $2(500)^{h/3}$

B. $500(2)^{3h}$

C. $500(3)^{h/2}$

D. $500(2)^{h/3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q4/5

A gym charges a $90 joining fee and $35 per month, so the average cost per month over m months is 35 + 90/m dollars. Explain what the part 90/m means.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSA-SSE.A.1.a

FORWARD 2

Rewrite an exponential expression to reveal a different rate • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b

Rewrite $12{,}500(0.97)^{t}$ (t in years) to show the monthly factor, rounded to 5 decimal places.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q1/5

$A(t) = 200(1/2)^{t/6}$. Write A as $200 \cdot b^t$ and give b to 4 decimal places. What is the hourly percent decrease?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q2/5

True or false?

$1.04^{n} = (1.04^{1/12})^{12n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q3/5

Which expression shows that $500(2)^{t/3}$ grows by about 26% per time unit?

A. $1.26(500)^{t}$

B. $500(1.26)^{3t}$

C. $500(1.26)^{t}$

D. $500(2.26)^{t}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S7-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q4/5

Explain why rewriting does not change the values of the function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.8.b • Set 7 • FORWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b • SEEING STRUCTURE IN EXPRESSIONS

Interpret complicated expressions by viewing parts as single entities

MAIN	Algebra II main question	**HSA-SSE.A.1.b**
BACKWARD 1	Squares are never negative	**8.EE.A.2**
BACKWARD 2	Powers of numbers greater than 1 and between 0 and 1	**8.EE.A.1**
BACKWARD 3	A fraction gets smaller as its denominator grows	**3.NF.A.3.d**
FORWARD 1	Find the set of all values of an expression on a domain	**HSA-SSE.A.1.b**
FORWARD 2	Use single-entity reasoning to find horizontal asymptotes	**HSF-IF.C.7.d**

SET 8 • HSA-SSE.A.1.b

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

$P(r) = 5000(1 + r)^{10}$ with $r > 0$ is the value of an investment. Without calculating, explain why $P(r) > 5000$.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q1/5

The height of a ball is $h(t) = 80 - 16(t - 2)^2$ feet after t seconds. Without expanding, find the greatest height and when it occurs, and explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q2/5

$C(x) = 10 + 40/(x + 1)$ is a cost in dollars for $x \geq 0$. Explain why $C(x)$ is always more than 10 and never more than 50.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q3/5

The temperature of a drink is $T(t) = 70 - 30(0.8)^t$ °F after t minutes. Explain why T(t) is always below 70 and gets closer to 70 as t grows.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q4/5

$f(x) = 3/(x^2 + 1)$. Find the greatest value of $f(x)$ and explain why no input gives a greater value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • MAIN • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 1

Squares are never negative • Grade 8 • Expressions & Equations • 8.EE.A.2

True or false?

$(t - 2)^2 \geq 0$ for every real t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q1/5

What is the smallest value of $(x + 5)^2$, and for which x?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q2/5

Which expression can be negative for some real x?

A. $x^2 + 1$

B. $x - 1$

C. x^2

D. $(x - 1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q3/5

What is the smallest value of $x^2 + 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q4/5

True or false?

$-(x - 3)^2$ is never positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 2

Powers of numbers greater than 1 and between 0 and 1 • Grade 8 • Expressions & Equations • 8.EE.A.1

Is 1.05^{10} greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q1/5

Is 0.8^5 greater than or less than 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q2/5

As n increases, 0.5^n

A. becomes negative

B. stays 0.5

C. gets closer to 0

D. grows without bound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q3/5

True or false?

A negative number raised to any whole-number power is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q4/5

Order from least to greatest: 0.9^2, 0.9^5, 0.9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSA-SSE.A.1.b

BACKWARD 3

A fraction gets smaller as its denominator grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

Which is greater, 40/2 or 40/5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q1/5

True or false?

If $x \geq 0$, then $40/(x + 1) \leq 40$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q2/5

As x grows, 6/(x + 3)

A. increases

B. becomes negative

C. decreases toward 0

D. stays 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q3/5

What is the greatest value of 12/(x + 4) for x ≥ 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q4/5

True or false?

8/10 is greater than 8/9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 1

Find the set of all values of an expression on a domain • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

Find every value that $h(t) = 80 - 16(t - 2)^2$ takes for $0 \le t \le 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q1/5

Find every value of $f(x) = 3/(x^2 + 1)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q2/5

True or false?

For $x \geq 0$, $C(x) = 10 + 40/(x + 1)$ takes every value in $10 < C \leq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q3/5

For $t \geq 0$, which set contains every value of $70 - 30(0.8)^t$?

A. $T \geq 70$

B. $40 \leq T < 70$

C. $40 < T \leq 70$

D. $0 < T < 70$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S8-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q4/5

Find every value of $g(x) = 5 - 2/(x^2 + 4)$ for real x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSA-SSE.A.1.b

FORWARD 2

Use single-entity reasoning to find horizontal asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Explain why $y = 10 + 40/(x + 1)$ has the horizontal asymptote $y = 10$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q1/5

Rewrite (2x + 7)/(x + 3) as 2 + 1/(x + 3) and state its horizontal asymptote.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q2/5

True or false?

$y = 3/(x^2 + 1)$ has the horizontal asymptote $y = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q3/5

The horizontal asymptote of $y = 5 - 2/(x - 4)$ is

A. $y = -2$

B. $y = 5$

C. $y = 4$

D. $y = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q4/5

Does the graph of $y = 10 + 40/(x + 1)$ ever reach its horizontal asymptote? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 8 • FORWARD 2 • Q5/5

Factor higher-degree expressions by recognizing differences of squares and sums or differences of cubes

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Take out a greatest common factor	**6.EE.A.3**
BACKWARD 2	Recognize perfect squares and perfect cubes	**8.EE.A.2**
BACKWARD 3	Multiply to check a factoring pattern	**HSA-APR.A.1**
FORWARD 1	Factor expressions whose parts are themselves expressions	**HSA-SSE.A.2**
FORWARD 2	Complete the factorization over the complex numbers	**HSN-CN.C.8**

SET 9 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 81$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q1/5

Factor $8x^3 + 27$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q2/5

Factor $x^6 - 64$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q3/5

Factor $16x^4 - y^4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q4/5

Factor $2x^5 - 32x$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • MAIN • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 1

Take out a greatest common factor • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor 6x + 18.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q1/5

Factor $2x^5 - 32x$ by taking out the greatest common factor only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q2/5

The greatest common factor of $12x^3$ and $18x$ is

A. $6x^3$

B. $6x$

C. $3x$

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q3/5

True or false?

$3x^2 + 9x = 3x(x + 9)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q4/5

Factor $5y^2 - 20y$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 2

Recognize perfect squares and perfect cubes • Grade 8 • Expressions & Equations • 8.EE.A.2

Write $16x^4$ as a square.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q1/5

Write $8x^3$ as a cube.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q2/5

Which is a perfect cube?

A. 32

B. 16

C. 36

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q3/5

True or false?

x^6 is both a perfect square and a perfect cube.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q4/5

Find the cube root of 27.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSA-SSE.A.2

BACKWARD 3

Multiply to check a factoring pattern • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(a + b)(a^2 - ab + b^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q1/5

Multiply $(x - 2)(x^2 + 2x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q2/5

True or false?

$(x + 3)(x^2 - 6x + 9) = x^3 + 27.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q3/5

$(x^2 - 9)(x^2 + 9)$ equals

A. $x^4 - 18x^2 - 81$

B. $x^4 + 81$

C. $(x - 3)^4$

D. $x^4 - 81$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-B3-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q4/5

Multiply $(2x + 1)(4x^2 - 2x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 1

Factor expressions whose parts are themselves expressions • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $(x + 1)^2 - 9$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q1/5

Factor $(2x - 1)^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q2/5

True or false?

$a^6 - b^6$ can be factored first as a difference of squares or first as a difference of cubes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q3/5

$(x - y)^2 - (x + y)^2$ equals

A. 0

B. $4xy$

C. $-4xy$

D. $-2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S9-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q4/5

Factor $x^2(x + 2) - 4(x + 2)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSA-SSE.A.2

FORWARD 2

Complete the factorization over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

Factor $x^4 - 81$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q1/5

Factor $16x^4 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q2/5

True or false?

Over the complex numbers, $x^3 + 8$ has only one linear factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q3/5

Which polynomial is fully factored over the integers but not over the complex numbers?

A. $x^2 + 9$

B. $x - 9$

C. $x^2 - 9$

D. $(x - 3)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q4/5

$2x^5 - 32x = 2x(x^2 + 4)(x + 2)(x - 2)$ over the integers.
Complete the factorization over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 9 • FORWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2 • SEEING STRUCTURE IN EXPRESSIONS

Rewrite expressions in quadratic form and factor

MAIN	Algebra II main question	**HSA-SSE.A.2**
BACKWARD 1	Factor quadratic trinomials	**HSA-SSE.A.2**
BACKWARD 2	Write powers as squares of other powers	**8.EE.A.1**
BACKWARD 3	Treat a repeated part as a single variable	**HSA-SSE.A.1.b**
FORWARD 1	Solve equations in quadratic form over the complex numbers	**HSN-CN.C.7**
FORWARD 2	Solve exponential equations in quadratic form	**HSF-LE.A.4**

SET 10 • HSA-SSE.A.2

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^4 - 5x^2 + 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q1/5

Factor $x^4 + 3x^2 - 10$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q2/5

Factor $9^x - 4 \cdot 3^x + 3$ by writing it in quadratic form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q3/5

Factor $x^6 - 9x^3 + 8$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q4/5

Factor $2x^4 - 7x^2 - 4$ completely over the integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • MAIN • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 1

Factor quadratic trinomials • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $u^2 - 5u + 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q1/5

Factor $u^2 + 3u - 10$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q2/5

$2u^2 - 7u - 4$ equals

A. $(2u + 1)(u - 4)$

B. $(u - 2)(2u + 2)$

C. $(2u + 4)(u - 1)$

D. $(2u - 1)(u + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

$u^2 - 9u + 8 = (u + 1)(u - 8)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q4/5

Factor $u^2 - 4u + 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 2

Write powers as squares of other powers • Grade 8 • Expressions & Equations • 8.EE.A.1

Write x^4 as a power of x^2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q1/5

Write 9^x as a power of 3^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q2/5

x^6 equals

A. $(x^4)^2$

B. $(x^3)^2$

C. $x^3 \cdot x^3 \cdot x^3$

D. $(x^2)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q3/5

True or false?

$4^x = (2^x)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q4/5

Write 25^x as a power of 5^x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSA-SSE.A.2

BACKWARD 3

Treat a repeated part as a single variable • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b

In $x^4 - 5x^2 + 4$, what single variable u turns the expression into $u^2 - 5u + 4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q1/5

Rewrite $2(x - 1)^2 + 3(x - 1) - 5$ using $u = x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q2/5

True or false?

In $4^x - 2^x - 6$, the substitution $u = 2^x$ gives $u^2 - u - 6$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q3/5

Which substitution writes $x^6 - 9x^3 + 8$ as a quadratic?

A. $u = x^2$

B. $u = x^3$

C. $u = x^6$

D. $u = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q4/5

After factoring $(u - 1)(u - 8)$ with $u = x^3$, write the result in terms of x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.b • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 1

Solve equations in quadratic form over the complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7

Solve $x^4 + 3x^2 - 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q1/5

Solve $x^4 - 5x^2 + 4 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q2/5

True or false?

$x^4 + 10x^2 + 9 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q3/5

How many complex solutions, with multiplicity, does $x^4 - 13x^2 + 36 = 0$ have?

A. 1

B. 3

C. 4

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q4/5

Solve $x^4 + x^2 - 12 = 0$ over the complex numbers.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.C.7 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSA-SSE.A.2

FORWARD 2

Solve exponential equations in quadratic form • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $9^x - 5 \cdot 3^x + 6 = 0$. Give exact solutions.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q1/5

Solve $4^x - 6 \cdot 2^x + 8 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q2/5

True or false?

$4^x + 2^x - 2 = 0$ has two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q3/5

The solutions of $25^x - 26 \cdot 5^x + 25 = 0$ are

A. 0 and 2

B. 1 and 25

C. 0 and 1

D. 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q4/5

Solve $e^{2t} - 7e^{t} + 10 = 0$. Give exact solutions and decimal values to 3 decimal places.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 10 • FORWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4 • SEEING STRUCTURE IN EXPRESSIONS

Derive the formula for the sum of a finite geometric series

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Write terms of a geometric sequence	**HSF-LE.A.2**
BACKWARD 2	Distribute a factor over a sum	**6.EE.A.3**
BACKWARD 3	Use the product rule for exponents	**8.EE.A.1**
FORWARD 1	Prove the sum formula from a polynomial identity	**HSA-APR.C.4**
FORWARD 2	Describe how partial sums behave when the ratio is between -1 a...	**HSA-SSE.B.4**

SET 11 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

$S_n = a + ar + ar^2 + \ldots + ar^{n-1}$ with $r \neq 1$. In the derivation of $S_n = a(1 - r^n)/(1 - r)$, which equation results from subtracting rS_n from S_n?

A. $S_n - rS_n = a - ar^n$

B. $S_n - rS_n = ar - ar^n$

C. $S_n - rS_n = a - ar$

D. $S_n - rS_n = a - ar^{n-1}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q1/5

$S = 3 + 6 + 12 + \ldots + 3 \cdot 2^9$. With the multiply-and-subtract method, $2S - S$ equals which expression, and what is S?

A. $3 \cdot 2^9 - 3$, so S = 1533

B. $3 \cdot 2^{10} + 3$, so S = 3075

C. $3 \cdot 2^{10} - 6$, so S = 3066

D. $3 \cdot 2^{10} - 3$, so S = 3069

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q2/5

Why does the derivation of $S_n = a(1 - r^n)/(1 - r)$ fail when $r = 1$, and what is S_n then?

A. When $r = 1$ the terms a, ar, ar^2, ... grow without bound, so S_n does not exist and no formula for it is possible.

B. Its last step divides by $1 - r$, which is 0 when $r = 1$. Every term then equals a, so $S_n = na$.

C. rS_n cannot be formed when $r = 1$, so $S_n = a$.

D. The formula gives 0/0, which means $S_n = 0$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q3/5

$T = 1 + x + x^2 + \ldots + x^n$ ($n + 1$ terms, $x \neq 1$).
Multiplying by x and subtracting T gives which formula?
(Check: $x = 2$, $n = 3$ gives $1 + 2 + 4 + 8 = 15$.)

A. $T = (x^n - 1)/(x - 1)$

B. $T = (x^{n+1} - x)/(x - 1)$

C. $T = (x^{n+1} - 1)/(x - 1)$

D. $T = (x^{n+1} - 1)/(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q4/5

$T = 100 + 100(1.05) + 100(1.05)^2 + \ldots + 100(1.05)^{19}$ (20 yearly amounts). Using the multiply-and-subtract method with $1.05T - T$, which equation and total are correct? A calculator may be used.

A. $0.05T = 100(1.05^{19} - 1)$, so $T \approx \$3{,}053.90$

B. $1.05T = 100(1.05^{20} - 1)$, so $T \approx \$157.46$

C. $0.05T = 100(1.05^{21} - 1)$, so $T \approx \$3{,}571.93$

D. $0.05T = 100(1.05^{20} - 1)$, so $T \approx \$3{,}306.60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • MAIN • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 1

Write terms of a geometric sequence • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the first four terms of the geometric sequence with first term 3 and ratio 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q1/5

What is the 10th term of 3, 6, 12, ...?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q2/5

The nth term of a geometric sequence with first term a and ratio r is

A. $a + (n - 1)r$

B. $a \cdot n \cdot r$

C. ar^n

D. ar^{n-1}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-B1-Q3**

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q3/5

True or false?

100, 105, 110.25, ... is arithmetic with common difference 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q4/5

Find the ratio of 81, 27, 9, 3, ...

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 2

Distribute a factor over a sum • Grade 6 • Expressions & Equations • 6.EE.A.3

Distribute: 2(3 + 6 + 12).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q1/5

Distribute: $r(a + ar + ar^2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q2/5

True or false?

$x(1 + x + x^2) = x + x^2 + x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q3/5

1.05(100 + 105) equals

A. 105 + 105

B. 100 + 110.25

C. 105 + 110.25

D. 205

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q4/5

Subtract: $(6 + 12 + 24 + 48) - (3 + 6 + 12 + 24)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSA-SSE.B.4

BACKWARD 3

Use the product rule for exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Simplify $r \cdot r^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q1/5

Simplify $ar^{n-1} \cdot r$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q2/5

$2 \cdot 2^9$ equals

A. 2^{10}

B. 4^{10}

C. 4^9

D. 2^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q3/5

True or false?

$x \cdot x^n = x^{2n}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q4/5

Evaluate 2^{10}.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 11 • BACKWARD 3 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 1

Prove the sum formula from a polynomial identity • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity $x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \ldots + x + 1)$?

A. Divide both sides by $x - 1$ to get $(x^n - 1)/(x - 1) = x^{n-1} + \ldots + x + 1$; this is the geometric sum formula, so the identity is true.

B. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and -1, leaving $x^n - 1$.

C. x times the sum is $x^n + x^{n-1} + \ldots + x$; subtracting the sum $x^{n-1} + \ldots + x + 1$ cancels every term except x^n and 1, leaving $x^n + 1$.

D. For $x = 2$ and $n = 3$: $7 = 1 \cdot 7$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S11-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q1/5

Use the identity $x^n - 1 = (x - 1)(x^{n-1} + \ldots + 1)$ to write $1 + 3 + 9 + \ldots + 3^5$ as a quotient, and evaluate it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q2/5

True or false?

The identity shows that $x - 1$ is a factor of $x^n - 1$ for every whole number $n \geq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q3/5

Which equals $1 + 2 + 4 + \ldots + 2^9$?

A. 2^{10}

B. $2^{10} - 1$

C. $2^9 - 1$

D. $(2^9 - 1)/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q4/5

Explain how the identity gives $a(1 - r^n)/(1 - r)$ for $a + ar + \ldots + ar^{n-1}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSA-SSE.B.4

FORWARD 2

Describe how partial sums behave when the ratio is between −1 and 1 • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

For the series $8 + 4 + 2 + 1 + \ldots$, find S_5 and S_{10} to 4 decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q1/5

Using $S_n = 16(1 - 0.5^n)$, explain why every partial sum of 8 + 4 + 2 + ... is less than 16 but can be made as close to 16 as you like.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q2/5

True or false?

For r = 2, the partial sums of $a + ar + ar^2 + \ldots$ ($a > 0$) stay below a fixed number.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q3/5

For $10 - 5 + 2.5 - 1.25 + \ldots$, the partial sums get close to

A. 20/3

B. 15

C. 5

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q4/5

A ball travels 10 m down, then rebounds and falls in a geometric pattern so the total distance after n bounces is $10 + 2 \cdot 6(1 - 0.6^n)/0.4$. What total can the distance never reach, and why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 11 • FORWARD 2 • Q5/5

Use the finite geometric series formula to solve problems

MAIN	Algebra II main question	**HSA-SSE.B.4**
BACKWARD 1	Identify the first term, ratio, and number of terms	**HSF-LE.A.2**
BACKWARD 2	Change a percent to a growth factor	**7.RP.A.3**
BACKWARD 3	Round results of calculations correctly	**HSN-Q.A.3**
FORWARD 1	Find the number of terms with logarithms	**HSF-LE.A.4**
FORWARD 2	Compare two payment plans with the series formula	**HSA-SSE.B.4**

SET 12 • HSA-SSE.B.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Find the sum of the first 12 terms of 5 + 15 + 45 + ...

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q1/5

Leah deposits $250 at the end of each month into an account that earns 0.5% interest per month. The balance just after the 36th deposit is $250 + 250(1.005) + \ldots + 250(1.005)^{35}$. Find it to the nearest cent.

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q2/5

A $180,000 loan is repaid with 360 equal monthly payments P at a monthly interest rate of 0.4%. P satisfies
$180{,}000 = P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$. Use the geometric series formula to find P to the nearest cent.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q3/5

A ball is dropped from 10 m and each rebound reaches 60% of the previous height. Find the total vertical distance it has traveled when it hits the ground for the 6th time, to the nearest hundredth of a meter.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q4/5

What is the least number of terms of 2 + 6 + 18 + ... whose sum is greater than 10,000? Show the sums that decide it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • MAIN • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 1

Identify the first term, ratio, and number of terms • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

For 5 + 15 + 45 + ... (12 terms), give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q1/5

For $250 + 250(1.005) + \ldots + 250(1.005)^{35}$, give a, r, and n.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q2/5

For 6 + 3.6 + 2.16 + 1.296 + 0.7776, r is

A. 0.4

B. 1.67

C. 0.6

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q3/5

True or false?

A list with exponents 0 through 35 has 35 terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q4/5

For $P/1.004 + P/1.004^2 + \ldots + P/1.004^{360}$, give the first term and the ratio.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 2

Change a percent to a growth factor • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

What factor adds 0.5% interest?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q1/5

A height is 60% of the one before. What is the ratio?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q2/5

0.4% per month as a factor is

A. 1.04

B. 1.004

C. 0.004

D. 1.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q3/5

True or false?

Interest of 12% per year is a factor of 0.12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q4/5

A loss of 15% per year is what factor?

Factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSA-SSE.B.4

BACKWARD 3

Round results of calculations correctly • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3

Round 9834.0262 to the nearest cent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q1/5

Round 37.6672 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q2/5

True or false?

Rounding 1.005 to 1.01 before raising it to the 36th power gives the same answer to the nearest cent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q3/5

1.005^{36} is closest to

A. 1.1967

B. 1.18

C. 1.2

D. 1.036

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q4/5

Evaluate 3^{12} exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.3 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 1

Find the number of terms with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

How many end-of-month deposits of $250 at 0.5% per month are needed for the balance to reach $20,000? Use $250(1.005^n - 1)/0.005 \geq 20{,}000$.

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q1/5

Solve $2(3^n - 1)/2 = 3^n - 1 = 1{,}000{,}000$ for n exactly, then give the least whole number of terms whose sum is at least 1,000,000.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q2/5

True or false?

For a ratio greater than 1, more terms always give a larger sum when $a > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q3/5

The sum $5(2^n - 1)$ first exceeds 5,000 when n =

A. 9

B. 11

C. 10

D. 1000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q4/5

Explain why the number of terms must be rounded up, not to the nearest whole number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSA-SSE.B.4

FORWARD 2

Compare two payment plans with the series formula • High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4

Plan A pays $1,000 per month for 24 months. Plan B pays $500 the first month and 8% more each month after that, for 24 months. Which plan pays more in total, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q1/5

Plan A pays $1,000 per month. Plan B pays $500 the first month and 8% more each month after that. After how many months does Plan B's total first exceed Plan A's total for the same number of months?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q2/5

True or false?

A geometric plan with ratio greater than 1 eventually pays more in total than any plan with a fixed monthly amount.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q3/5

Plan C pays $2,000 then 5% less each month. Its total over 24 months is closest to

A. $40,000

B. $24,000

C. $48,000

D. $28,320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q4/5

Name one assumption in comparing the plans by their totals only.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Algebra II • Algebra: Seeing Structure in Expressions • HSA-SSE.B.4 • Set 12 • FORWARD 2 • Q5/5

Add and subtract polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Combine like terms with the same power	**7.EE.A.1**
BACKWARD 2	Subtract a whole expression	**6.EE.A.3**
BACKWARD 3	Write polynomials in standard form	**HSA-SSE.A.1.a**
FORWARD 1	Reason about the degree of a sum or difference	**HSA-APR.A.1**
FORWARD 2	Add and subtract rational expressions	**HSA-APR.D.7**

SET 13 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Simplify $(4x^3 - 2x^2 + 7) + (-x^3 + 5x - 9)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q1/5

Simplify $(6x^4 - x^3 + 2x) - (2x^4 + 3x^3 - x^2 + 2x - 1)$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q2/5

A company's revenue is $R(x) = -0.01x^3 + 3x^2 + 40x$ dollars and its cost is $C(x) = 0.02x^2 + 15x + 900$ dollars for x items. Write the profit $P(x) = R(x) - C(x)$ in standard form.

$P(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q3/5

Simplify $(x^5 - 3x^3 + x) - (x^5 + 2x^4 - 3x^3 - 7)$. Write the result in standard form and give its degree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q4/5

$p(x) = 5x^3 + x^2 - 6$. Find the polynomial $q(x)$ for which $p(x) + q(x) = 2x^3 - x + 4$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 13 • MAIN • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 1

Combine like terms with the same power • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify $4x^3 - x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q1/5

Simplify $-2x^2 + 5x - 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q2/5

True or false?

$x^3 + x^2 = x^5$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q3/5

Which pair are like terms?

A. $4x^2$ and $-x^2$

B. $4x^2$ and $4x$

C. x^3 and $3x$

D. x and 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q4/5

Simplify $7 - 9 + 2x - 2x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 2

Subtract a whole expression • Grade 6 • Expressions & Equations • 6.EE.A.3

Rewrite $-(2x^4 + 3x^3 - x^2 + 2x - 1)$ without parentheses.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q1/5

Simplify $(5 - x) - (3 - 4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q2/5

True or false?

$a - (b - c) = a - b - c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q3/5

$-(x^2 - 7)$ equals

A. $x^2 + 7$

B. $-x^2 - 7$

C. $-x^2 + 7$

D. $x^2 - 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q4/5

Simplify $10 - (x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSA-APR.A.1

BACKWARD 3

Write polynomials in standard form • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a

Write $7 + 5x - 2x^3 + x^2$ in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q1/5

What is the leading coefficient of $3 - 4x^2 + x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q2/5

The degree of $-2x^4 + x + 7$ is

A. 7

B. 1

C. 4

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q3/5

True or false?

The standard form of $2x - x^3$ is $2x - x^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q4/5

What is the constant term of $6x^3 - 2x + 9$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.1.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 1

Reason about the degree of a sum or difference • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

p(x) has degree 4 and q(x) has degree 4. What degrees can p(x) + q(x) have? Give an example where the degree drops.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q1/5

For which value of k does
$(kx^3 + 2x - 1) - (5x^3 - x^2 + 3)$ have degree 2?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q2/5

True or false?

The difference of two polynomials is always a polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q3/5

p has degree 5 and q has degree 3. The degree of $p - q$ is

A. 5

B. 2

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q4/5

Explain why the sum of two polynomials is a polynomial.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSA-APR.A.1

FORWARD 2

Add and subtract rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Write 3/x + 2/(x + 1) as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q1/5

Write $\frac{1}{(x - 2)} - \frac{1}{(x + 2)}$ as one rational expression, and state the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q2/5

True or false?

Polynomials are closed under division.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q3/5

x/2 + x/3 equals

A. 5x/5

B. x/5

C. 5x/6

D. 2x/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S13-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q4/5

Explain how adding rational expressions is like adding fractions of integers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 13 • FORWARD 2 • Q5/5

Multiply polynomials of higher degree

MAIN	Algebra II main question	**HSA-APR.A.1**
BACKWARD 1	Distribute a monomial over a polynomial	**HSA-APR.A.1**
BACKWARD 2	Multiply powers with the same base	**8.EE.A.1**
BACKWARD 3	Multiply two binomials	**HSA-APR.A.1**
FORWARD 1	Find a coefficient of a product without full expansion	**HSA-APR.A.1**
FORWARD 2	Expand powers of binomials with the Binomial Theorem	**HSA-APR.C.5**

SET 14 • HSA-APR.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x^2 + 3x - 2)(2x - 5)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q1/5

Expand $(x - 2)^3$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q2/5

Multiply $(x^2 - x + 1)(x^2 + x + 1)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q3/5

Multiply $(2x + 1)(x - 3)(x + 4)$. Write the product in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q4/5

A box has length $x + 2$, width $x - 1$, and height $2x + 3$ (in cm). Write its volume as a polynomial in standard form.

$V(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • MAIN • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 1

Distribute a monomial over a polynomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $2x(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q1/5

Multiply $-5(x^2 + 3x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q2/5

True or false?

$x^2(x + 4) = x^3 + 4x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q3/5

$-3x(x - 1)$ equals

A. $3x^2 - 3x$

B. $-3x^2 + 3x$

C. $-3x^2 - 3x$

D. $-3x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-B1-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q4/5

Multiply $x(x^2 - x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 2

Multiply powers with the same base • Grade 8 • Expressions & Equations • 8.EE.A.1

Simplify $x^2 \cdot x^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q1/5

Simplify $2x \cdot 3x^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$x^2 \cdot x^2 = x^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q3/5

$-x \cdot x^3$ equals

A. x^4

B. $-x^3$

C. $-x^4$

D. $-x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q4/5

Simplify $(x^2)(-4x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSA-APR.A.1

BACKWARD 3

Multiply two binomials • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Multiply $(x - 3)(x + 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q1/5

Multiply $(x + 2)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q2/5

$(x - 2)^2$ equals

A. $x^2 + 4$

B. $x^2 - 4$

C. $x^2 - 4x + 4$

D. $x^2 - 2x + 4$

True or false?

$(2x + 1)(x - 3) = 2x^2 - 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q4/5

Multiply $(2x + 3)(x - 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 1

Find a coefficient of a product without full expansion • High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find the coefficient of x^2 in $(x^2 + 3x - 2)(4x^2 - x + 5)$ without expanding everything.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q1/5

For which k does $(x + k)(x^2 - 3x + 1)$ have no x^2 term?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q2/5

True or false?

The constant term of a product of polynomials is the product of their constant terms.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q3/5

The leading coefficient of $(2x^3 - 1)(5x^2 + x)$ is

A. 7

B. 5

C. 2

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q4/5

Find the coefficient of x in $(x - 1)(x - 2)(x - 3)$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.A.1 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSA-APR.A.1

FORWARD 2

Expand powers of binomials with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Expand $(x + 1)^4$ using Pascal's Triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q1/5

Expand $(x - 2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

The coefficients of $(x + y)^5$ are 1, 5, 10, 5, 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q3/5

The coefficient of x^2y^2 in $(x + y)^4$ is

A. 2

B. 6

C. 4

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q4/5

Explain why the coefficients of $(x + y)^n$ are symmetric.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 14 • FORWARD 2 • Q5/5

Find remainders with the Remainder Theorem

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Evaluate a polynomial function	**HSF-IF.A.2**
BACKWARD 2	Powers of negative numbers	**7.NS.A.2**
BACKWARD 3	Dividend = divisor × quotient + remainder	**4.NBT.B.6**
FORWARD 1	Find unknown coefficients from given remainders	**HSA-APR.B.2**
FORWARD 2	Prove the Remainder Theorem	**HSA-APR.B.2**

SET 15 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Use the Remainder Theorem to find the remainder when $p(x) = x^3 - 4x^2 + 2x - 5$ is divided by $x - 3$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q1/5

Use the Remainder Theorem to find the remainder when $p(x) = 2x^4 + x^3 - 3x + 7$ is divided by $x + 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q2/5

Use the Remainder Theorem to find the remainder when $p(x) = x^{100} + 3$ is divided by $x + 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q3/5

Use the Remainder Theorem to find the remainder when $p(x) = 4x^3 - 6x^2 + x - 1$ is divided by $x - 1/2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q4/5

Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 5x^2 + 4x$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • MAIN • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 1

Evaluate a polynomial function • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$p(x) = x^3 - 4x^2 + 2x - 5$. Find $p(1)$.

$p(1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q1/5

$p(x) = 2x^2 - x + 3$. Find $p(-2)$.

$p(-2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q2/5

True or false?

If $p(x) = x^2 + 1$, then $p(-3) = -8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q3/5

If $p(x) = x^3 - x$, then $p(2) =$

A. 4

B. 6

C. 8

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q4/5

$p(x) = 4x - 1$. Find $p(1/2)$.

$p(1/2) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 2

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2

Evaluate $(-2)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q1/5

Evaluate $(-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q2/5

$(-1)^{100}$ equals

A. -100

B. -1

C. 100

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q3/5

True or false?

$-2^4 = (-2)^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q4/5

Evaluate $2(-2)^4 + (-2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSA-APR.B.2

BACKWARD 3

Dividend = divisor × quotient + remainder • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Write $47 = 5 \times q + r$ with $0 \leq r < 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

A division gives quotient 12 and remainder 3 with divisor 7. What was the dividend?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

If a division leaves remainder 0, the divisor is a factor of the dividend.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

$x^3 - 1 = (x - 1)(x^2 + x + 1) + 0$. The remainder on division by $x - 1$ is

A. 0

B. 1

C. $x^2 + x + 1$

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

$p(x) = (x - 3)(x^2 - x - 1) - 8$. What is the remainder when $p(x)$ is divided by $x - 3$?

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 1

Find unknown coefficients from given remainders • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

When $p(x) = x^3 + kx^2 - 4$ is divided by $x - 2$, the remainder is 12. Find k.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q1/5

$p(x) = x^3 + ax + b$ leaves remainder 5 on division by $x - 1$ and remainder -1 on division by $x + 1$. Find a and b.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q2/5

True or false?

If p(x) leaves remainder 0 on division by $x - 4$, then $p(-4) = 0$.

A. True

B. False

$p(x) = 2x^2 + kx + 1$ leaves remainder 7 on division by $x + 1$. Then $k =$

A. 4

B. -4

C. -6

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q4/5

For which k is the remainder of $x^4 - kx + 6$ divided by $x + 2$ equal to 0?

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSA-APR.B.2

FORWARD 2

Prove the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Which argument proves that the remainder when a polynomial $p(x)$ is divided by $x - a$ is $p(a)$?

A. Write $p(x) = (x - a)q(x) + r$, where r is a constant because it has degree less than 1. Substituting $x = a$ gives $p(a) = 0 \cdot q(a) + r = r$.

B. Dividing x^3 by $x - 2$ leaves remainder 8, and $2^3 = 8$, so the remainder is always $p(a)$.

C. Since $x - a = 0$ exactly when $x = a$, dividing $p(x)$ by $x - a$ is the same as replacing x with a, so the remainder must be $p(a)$.

D. Write $p(x) = (x - a)q(x) + r$. Substituting $x = 0$ gives $p(0) = (0 - a)q(0) + r = -a \cdot q(0) + r$, so the remainder r equals $p(a)$ for every a.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S15-F2-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q1/5

Why must the remainder on division by $x - a$ be a constant?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q2/5

True or false?

The Remainder Theorem also works for division by $2x - 1$ with remainder $p(1/2)$.

A. True

B. False

Which step uses the fact that $a - a = 0$?

A. Graphing $p(x)$

B. Dividing $p(x)$ by x

C. Factoring $p(x)$

D. Substituting $x = a$ into $p(x) = (x - a)q(x) + r$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q4/5

Explain how the proof shows that $p(a) = 0$ exactly when $x - a$ is a factor of $p(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 15 • FORWARD 2 • Q5/5

Use p(a) = 0 to find factors

MAIN	Algebra II main question	**HSA-APR.B.2**
BACKWARD 1	Use the Remainder Theorem	**HSA-APR.B.2**
BACKWARD 2	Factor the quadratic quotient	**HSA-SSE.A.2**
BACKWARD 3	Divide a polynomial by a linear factor	**HSA-APR.A.1**
FORWARD 1	Choose parameters so that a given binomial is a factor	**HSA-APR.B.2**
FORWARD 2	Finish factorizations over the complex numbers	**HSN-CN.C.8**

SET 16 • HSA-APR.B.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Show that $x - 2$ is a factor of $p(x) = x^3 - 3x^2 - 4x + 12$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q1/5

Show that $x + 1$ is a factor of $p(x) = x^3 + 2x^2 - 5x - 6$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q2/5

Show that $x + 3$ is a factor of $p(x) = 2x^3 + 5x^2 - 4x - 3$, then factor $p(x)$ completely.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q3/5

Is $x - 4$ a factor of $p(x) = x^3 - 5x^2 + 2x + 9$? Justify with the Remainder Theorem.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q4/5

$p(x) = x^3 - 7x + 6$. Use $p(a) = 0$ to find all the zeros of p, and write p(x) in factored form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • MAIN • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 1

Use the Remainder Theorem • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder when $x^3 - 3x^2 - 4x + 12$ is divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q1/5

Find the remainder when $x^3 + 2x^2 - 5x - 6$ is divided by $x - 1$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q2/5

True or false?

Dividing by $x + 3$ means evaluating at $x = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q3/5

The remainder of $p(x) \div (x + 1)$ is

A. $-p(1)$

B. $p(-1)$

C. $p(0)$

D. $p(1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q4/5

Find $p(-3)$ for $p(x) = 2x^3 + 5x^2 - 4x - 3$.

$p(-3) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 2

Factor the quadratic quotient • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q1/5

Factor $2x^2 - x - 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q2/5

$x^2 + x - 6$ equals

A. $(x - 3)(x + 2)$

B. $(x - 6)(x + 1)$

C. $(x + 3)(x - 2)$

D. $(x + 6)(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q3/5

True or false?

$x^2 + 4$ factors over the integers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSA-APR.B.2

BACKWARD 3

Divide a polynomial by a linear factor • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Find $q(x)$ if $x^3 - 3x^2 - 4x + 12 = (x - 2)\,q(x)$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q1/5

Multiply $(x + 1)(x^2 + x - 6)$ to check a division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q2/5

True or false?

If $(x - 1)(x^2 + x - 6) = x^3 - 7x + 6$, then $x^2 + x - 6$ is the quotient of $x^3 - 7x + 6 \div (x - 1)$.

A. True

B. False

$x^3 - 8 \div (x - 2)$ equals

A. $x^2 + 2x + 4$

B. $x^2 + 4$

C. $x^2 - 4$

D. $x^2 - 2x + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-B3-Q4**

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q4/5

Find $q(x)$ if $2x^3 + 5x^2 - 4x - 3 = (x + 3)\, q(x)$.

$q(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 1

Choose parameters so that a given binomial is a factor • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find k so that $x - 3$ is a factor of $x^3 + kx^2 - 3x + 9$.

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q1/5

Find a and b so that $x - 1$ and $x + 2$ are both factors of $x^3 + ax^2 + bx - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q2/5

True or false?

If $x - 2$ and $x + 2$ are factors of $p(x)$, then $x^2 + 4$ is a factor of $p(x)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q3/5

For which k is $x + 1$ a factor of $x^4 + kx + 3$?

A. 2

B. -4

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q4/5

Write a cubic with leading coefficient 1 that has $x - 1$, $x + 1$, and $x - 4$ as factors, in standard form.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSA-APR.B.2

FORWARD 2

Finish factorizations over the complex numbers • High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8

$x - 1$ is a factor of $x^3 - 1$. Factor $x^3 - 1$ completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q1/5

$x + 2$ is a factor of $x^3 + 2x^2 + 9x + 18$. Factor it completely over the complex numbers.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q2/5

True or false?

Every factor found with the factor theorem gives a zero of the polynomial.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q3/5

$x - 2$ is a factor of $x^3 - 2x^2 + x - 2$. The other zeros are

A. i and $-i$

B. -2

C. 1 and -1

D. $2i$ and $-2i$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S16-F2-Q4**

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q4/5

$x - 3$ is a factor of $x^3 - 3x^2 + 4x - 12$. List all complex zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Number & Quantity: The Complex Number System • HSN-CN.C.8 • Set 16 • FORWARD 2 • Q5/5

Identify zeros and multiplicities from a factorization

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Solve by the zero-product property	**HSA-REI.B.4.b**
BACKWARD 2	Factor to find hidden repeated factors	**HSA-SSE.A.2**
BACKWARD 3	An exponent counts repeated factors	**6.EE.A.1**
FORWARD 1	Build a polynomial from zeros and one more condition	**HSA-APR.B.3**
FORWARD 2	Decide what the zeros determine and what remains free	**HSA-APR.B.3**

SET 17 • HSA-APR.B.3

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $p(x) = (x - 4)^2(x + 1)(2x - 3)$ and give the multiplicity of each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q1/5

Factor $p(x) = x^3 - 6x^2 + 9x$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q2/5

List the zeros of $p(x) = (x^2 - 9)(x^2 + 2x + 1)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q3/5

List the zeros of $p(x) = -2x(x + 5)^3$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q4/5

Factor $p(x) = x^4 - 8x^2 + 16$, then list its zeros with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • MAIN • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 1

Solve by the zero-product property • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $(x - 4)(x + 1) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q1/5

Solve $x(2x - 3) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q2/5

True or false?

If $(x - 2)^2 = 0$, then $x = 2$ or $x = -2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q3/5

The solutions of $(x + 5)(x - 5) = 0$ are

A. 25

B. -5 and 5

C. 0 and 5

D. 5 only

Solve $-2x(x + 5) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 2

Factor to find hidden repeated factors • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 - 6x + 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q1/5

Factor $x^2 + 2x + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q2/5

$x^4 - 8x^2 + 16$ equals

A. $(x^2 + 4)^2$

B. $(x^2 - 4)^2$

C. $(x^2 - 8)^2$

D. $(x - 4)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q3/5

True or false?

$x^3 - 6x^2 + 9x = x(x + 3)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSA-APR.B.3

BACKWARD 3

An exponent counts repeated factors • Grade 6 • Expressions & Equations • 6.EE.A.1

Write $(x - 4)^2$ as a product without an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q1/5

How many factors of $(x + 5)$ are in $(x + 5)^3$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q2/5

True or false?

$(x - 2)^2(x - 2) = (x - 2)^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q3/5

In $7(x - 1)^4$, the factor $x - 1$ appears

A. 28 times

B. 4 times

C. 1 time

D. 7 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q4/5

Write $(x + 1)(x + 1)(x - 3)$ with an exponent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 1

Build a polynomial from zeros and one more condition • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Write the cubic polynomial with zeros -1, 2, and 4 (each multiplicity 1) whose graph passes through $(0, 16)$.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q1/5

Write a polynomial of degree 3 with a zero of multiplicity 2 at $x = 3$, a zero at $x = -2$, and leading coefficient -1.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q2/5

True or false?

Zeros alone determine a polynomial completely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q3/5

A quartic has zeros 1 and −1, each of multiplicity 2, and p(0) = 3. Then p(x) =

A. $-3(x-1)^2(x+1)^2$

B. $3(x^2-1)$

C. $(x-1)^2(x+1)^2$

D. $3(x-1)^2(x+1)^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q4/5

A polynomial has zeros 0 (multiplicity 2) and 5 (multiplicity 1), degree 3, and $p(1) = -8$. Find $p(x)$.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSA-APR.B.3

FORWARD 2

Decide what the zeros determine and what remains free • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

All polynomials with zeros 1 and 3 only (each multiplicity 1) and degree 2 have what form?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q1/5

If the degree is not fixed, give two different polynomials whose only real zeros are 1 and 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q2/5

True or false?

Two cubic polynomials with the same three zeros (each multiplicity 1) must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q3/5

Which extra information fixes a cubic with zeros −2, 0, 2?

A. That it has degree 3

B. That 0 is a zero

C. Its end behavior only

D. Its value at x = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q4/5

Explain why knowing only the real zeros of a degree-4 polynomial may not determine it even up to scale.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 17 • FORWARD 2 • Q5/5

Construct a rough graph of a polynomial from its zeros

MAIN	Algebra II main question	**HSA-APR.B.3**
BACKWARD 1	Read zeros and multiplicities	**HSA-APR.B.3**
BACKWARD 2	End behavior from the leading term	**HSF-IF.B.4**
BACKWARD 3	Find the y-intercept by evaluating at 0	**HSF-IF.A.2**
BACKWARD 4	Sign of a power of a negative or positive number	**7.NS.A.2**
FORWARD 1	Sketch graphs from standard form by factoring first	**HSF-IF.C.7.c**
FORWARD 2	Write a polynomial from its rough graph	**HSA-APR.B.3**

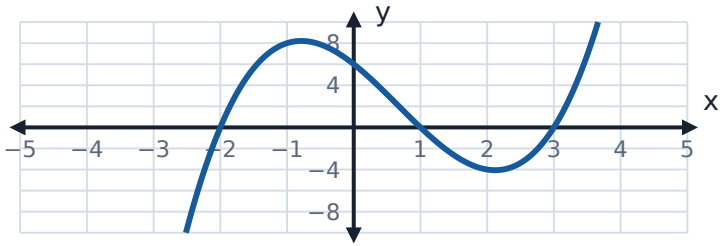
SET 18 • HSA-APR.B.3

MAIN

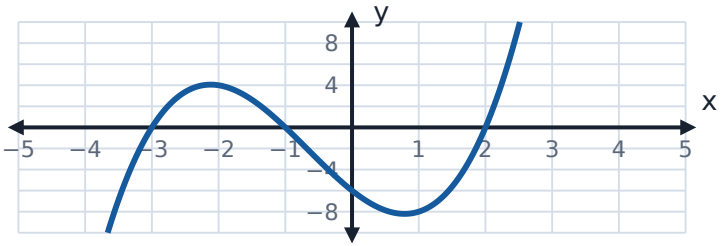
Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

Which graph is a rough graph of $y = (x + 2)(x - 1)(x - 3)$?

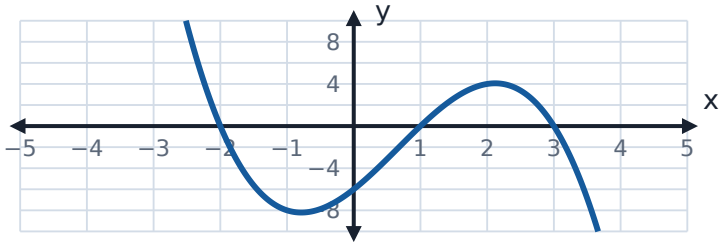
A



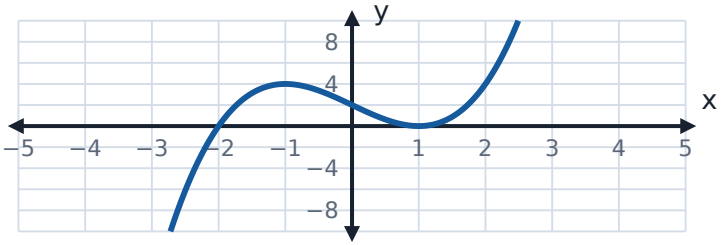
B



C



D



A. Graph A

B. Graph B

C. Graph C

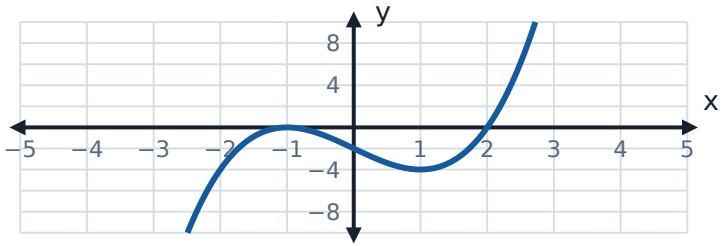
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q1**

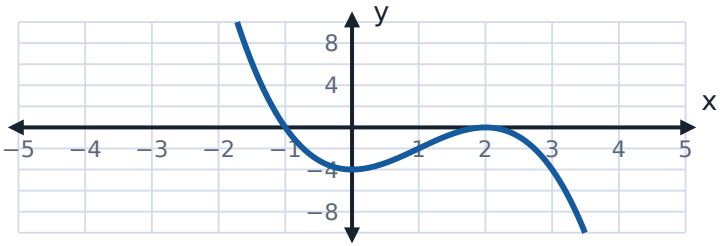
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q1/5

Which graph is a rough graph of $y = -(x + 1)^2(x - 2)$?

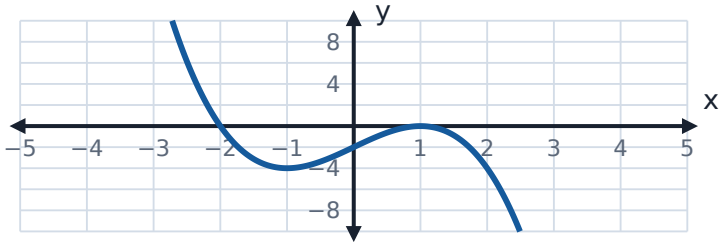
A



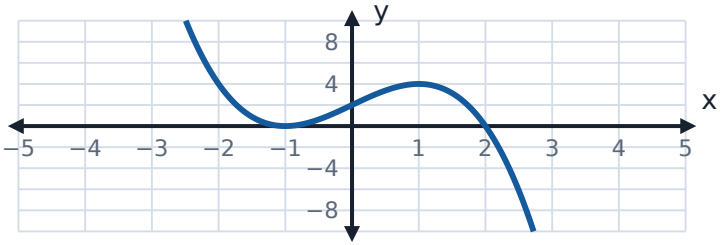
B



C



D



A. Graph A

B. Graph B

C. Graph C

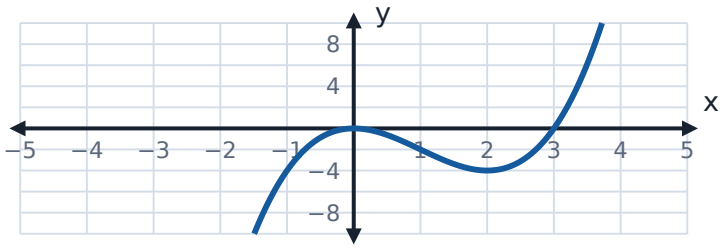
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q2**

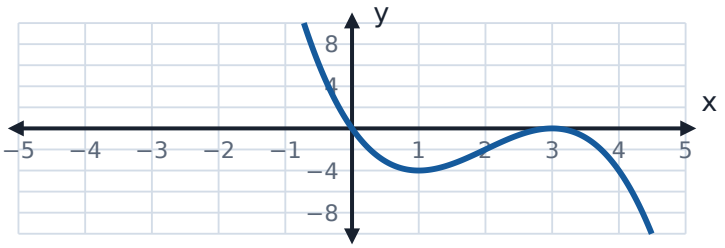
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q2/5

Which graph is a rough graph of $y = x(x - 3)^2$?

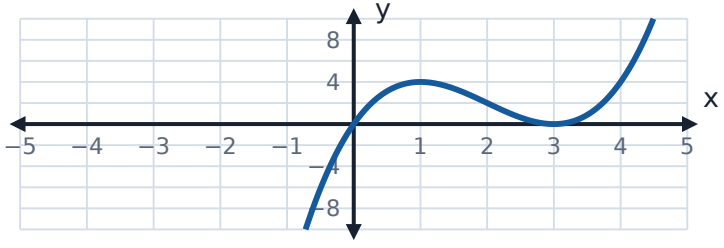
A



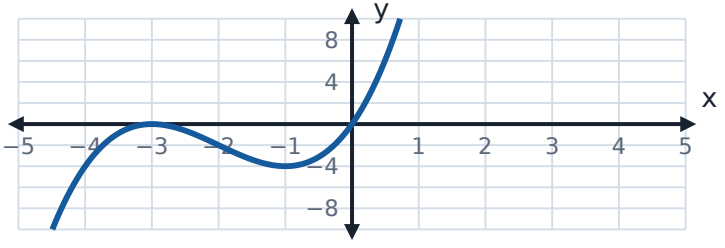
B



C



D



A. Graph A

B. Graph B

C. Graph C

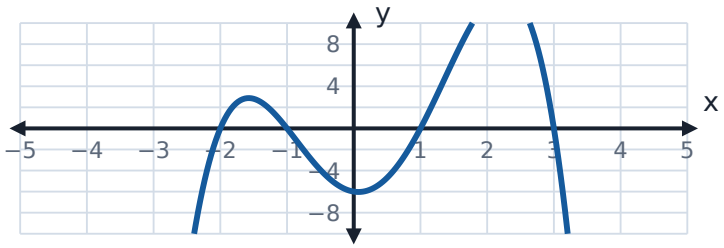
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q3**

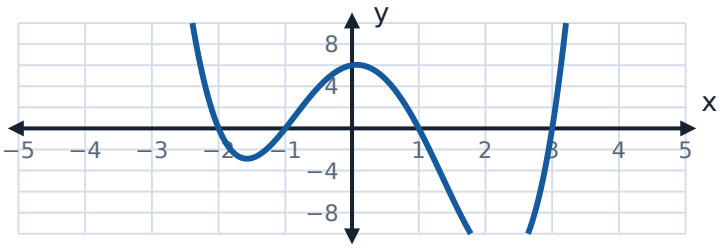
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q3/5

Which graph is a rough graph of $y = (x + 2)(x + 1)(x - 1)(x - 3)$?

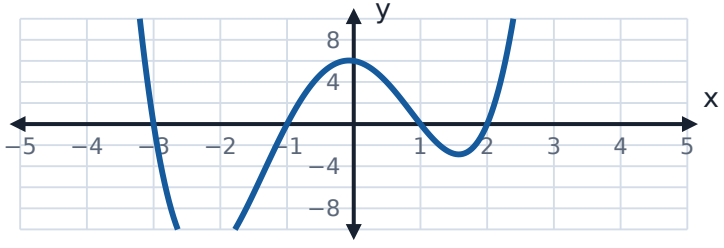
A



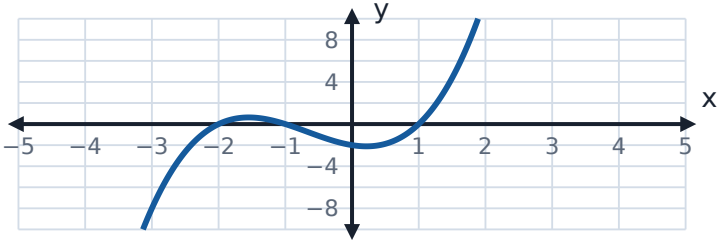
B



C



D



A. Graph A

B. Graph B

C. Graph C

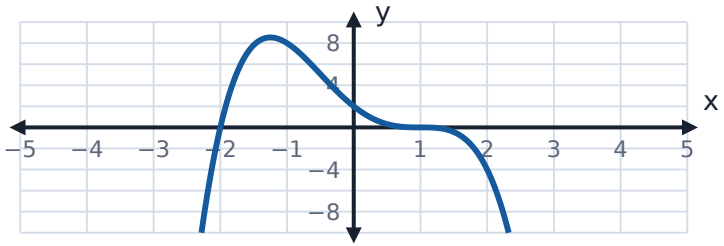
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q4**

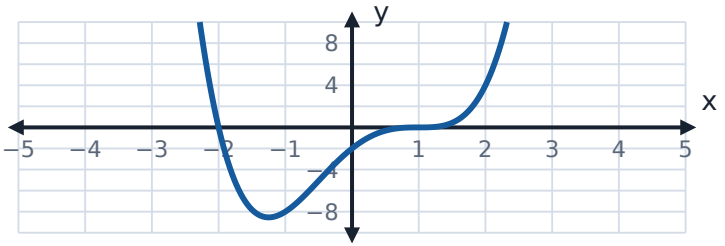
High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q4/5

Which graph is a rough graph of $y = -(x - 1)^3(x + 2)$?

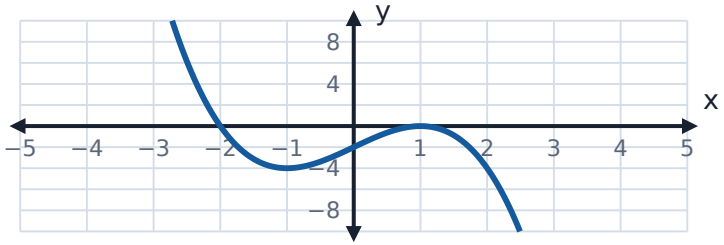
A



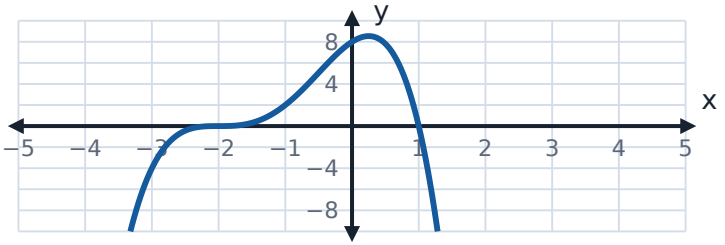
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-M-Q5**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • MAIN • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 1

Read zeros and multiplicities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q1/5

List the zeros of $-(x + 1)^2(x - 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q2/5

True or false?

$x(x - 3)^2$ has a zero of multiplicity 2 at $x = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q3/5

The zero $x = 1$ of $-(x - 1)^3(x + 2)$ has multiplicity

A. 1

B. 2

C. 4

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q4/5

List the zeros of $(x + 2)(x + 1)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 2

End behavior from the leading term • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

As x grows large and positive, what happens to x^3?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q1/5

As x grows large and negative, what happens to $-x^4$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q2/5

The leading term of $(x + 2)(x - 1)(x - 3)$ is

A. $-x^3$

B. -6

C. $3x$

D. x^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q3/5

True or false?

A polynomial of even degree with a positive leading coefficient falls at both ends.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q4/5

What is the leading term of $-(x-1)^3(x+2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 3

Find the y-intercept by evaluating at 0 • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find y when x = 0 for $y = (x + 2)(x - 1)(x - 3)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q1/5

Find y when x = 0 for $y = -(x + 1)^2(x - 2)$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q2/5

True or false?

The y-intercept of $x(x - 3)^2$ is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q3/5

The y-intercept of $-(x-1)^3(x+2)$ is

A. 0

B. -2

C. 2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q4/5

Find y when $x = 1$ for $y = x(x - 3)^2$.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 18 • BACKWARD 3 • Q5/5

SET 18 • HSA-APR.B.3

BACKWARD 4

Sign of a power of a negative or positive number • Grade 7 • The Number System • 7.NS.A.2

Is $(-0.5)^2$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q1/5

Is $(-0.5)^3$ positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q2/5

True or false?

An even power of a negative number is positive.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q3/5

Near $x = 3$, $(x - 3)^2$ is

A. negative on both sides

B. positive on both sides

C. zero everywhere

D. positive on the left and negative on the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q4/5

Does $(x - 1)^3$ change sign at $x = 1$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B4-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 18 • BACKWARD 4 • Q5/5

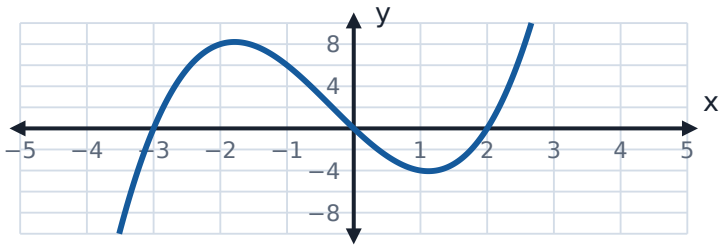
SET 18 • HSA-APR.B.3

FORWARD 1

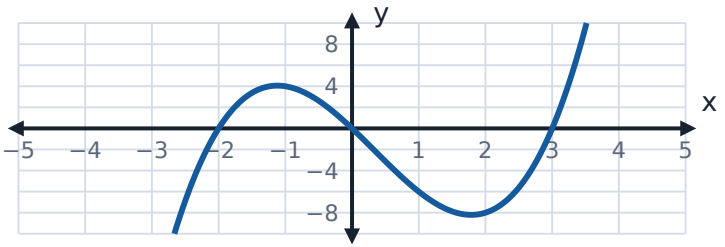
Sketch graphs from standard form by factoring first • High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c

Factor $y = x^3 - x^2 - 6x$. Which graph is a rough graph of it?

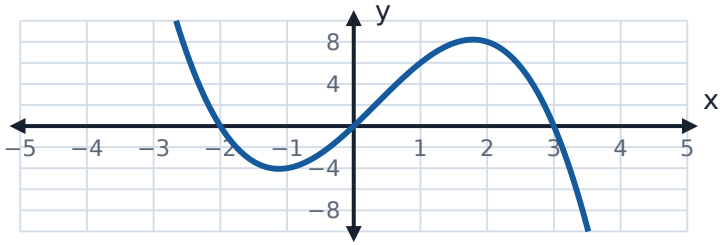
A



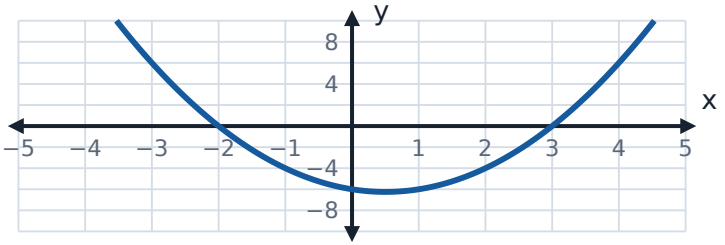
B



C



D



A. Graph A

B. Graph B

C. Graph C

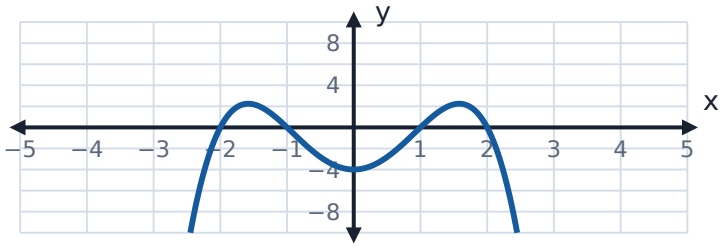
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q1**

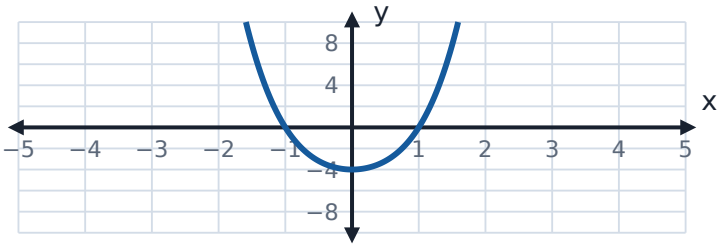
High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q1/5

Factor $y = x^4 - 5x^2 + 4$. Which graph is a rough graph of it?

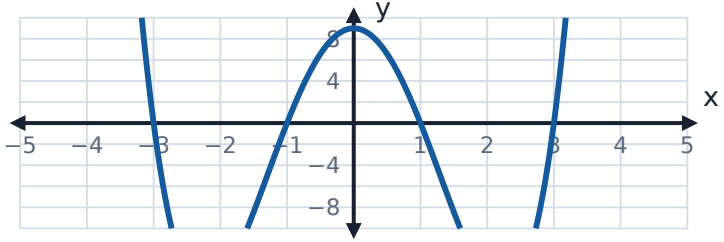
A



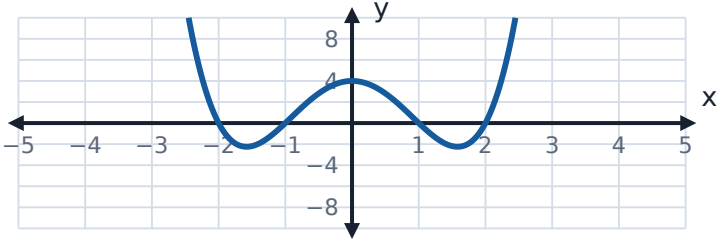
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F1-Q2**

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q2/5

True or false?

$y = x^3 + 4x$ crosses the x-axis at $x = -2$, 0, and 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q3/5

$y = -x^3 + 9x$ crosses the x-axis at

A. 0 and 9

B. 3 only

C. $-3, 0, 3$

D. $-9, 0, 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q4/5

$y = x^3 - 3x^2 + 4 = (x + 1)(x - 2)^2$. Describe its behavior at each zero.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School — Algebra II • Functions: Interpreting Functions • HSF-IF.C.7.c • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSA-APR.B.3

FORWARD 2

Write a polynomial from its rough graph • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

A cubic's graph crosses the x-axis at −3 and touches it at 2, and passes through (0, 12). Write the polynomial.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q1/5

A quartic rises at both ends, touches the x-axis at -1 and at 3, and passes through (0, 18). Write the polynomial.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q2/5

True or false?

A graph that touches the x-axis at x = a without crossing has a zero of even multiplicity at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q3/5

A cubic falls to the right and crosses the x-axis at 0, 1, and 4. Its leading coefficient is

A. zero

B. not determined by the end behavior

C. positive

D. negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S18-F2-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q4/5

A polynomial graph crosses at -2, touches at 1, falls to the left and rises to the right, and has y-intercept 4. Give a polynomial of least degree.

$p(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 18 • FORWARD 2 • Q5/5

Prove polynomial identities

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Expand the square of a binomial	**HSA-APR.A.1**
BACKWARD 2	Recognize a difference of squares	**HSA-SSE.A.2**
BACKWARD 3	What makes an identity	**6.EE.A.4**
FORWARD 1	Prove or disprove proposed identities	**HSA-APR.C.4**
FORWARD 2	Prove identities with the Binomial Theorem	**HSA-APR.C.5**

SET 19 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Which argument proves the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$?

A. For $x = 2$ and $y = 1$: $25 = 9 + 16$, so the identity is true.

B. Take the square root of both sides: $x^2 + y^2 = (x^2 - y^2) + 2xy$, and the right side simplifies to $x^2 - y^2 + 2xy$, so the identity is true.

C. The left side is $x^4 + 2x^2y^2 + y^4$. The right side is $x^4 - 2x^2y^2 + y^4 + 4x^2y^2 = x^4 + 2x^2y^2 + y^4$. The sides are equal.

D. The right side is $(x^2 - y^2)^2 + (2xy)^2 = x^4 - y^4 + 4x^2y^2$, and the left side is $x^4 + y^4 + 4x^2y^2$, so the two sides are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-M-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q1/5

Which argument proves the identity $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$?

A. The product is $a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3$. The terms $-a^2b$ and a^2b cancel, ab^2 and $-ab^2$ cancel, leaving $a^3 + b^3$.

B. For $a = 1$ and $b = 2$ the left side is $1 + 8 = 9$ and the right side is $3 \cdot (1 - 2 + 4) = 9$, so the identity is true for all a and b.

C. Multiply the first terms and the last terms: $a \cdot a^2 = a^3$ and $b \cdot b^2 = b^3$, so the product is $a^3 + b^3$.

D. The product is $a^3 - 2a^2b + b^3$, which equals $a^3 + b^3$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q2/5

Which argument proves the identity $(a + b)^2 - (a - b)^2 = 4ab$?

A. For $a = 3$ and $b = 1$: $16 - 4 = 12 = 4 \cdot 3$, so the identity is true.

B. $(a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = a^2 + 2ab + b^2 - a^2 + 2ab - b^2 = 4ab$.

C. $(a + b)^2 - (a - b)^2 = [(a + b) - (a - b)]^2 = (2b)^2 = 4ab$.

D. $(a + b)^2 - (a - b)^2 = (a^2 + b^2) - (a^2 - b^2) = a^2 + b^2 - a^2 + b^2 = 4ab$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q3/5

Which argument proves the identity $x^4 + 4 = (x^2 + 2x + 2)(x^2 - 2x + 2)$?

A. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 + (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

B. For $x = 0$: $4 = 2 \cdot 2$, so the identity is true.

C. The right side is $((x^2 + 2) + 2x)((x^2 + 2) - 2x) = (x^2 + 2)^2 - (2x)^2 = x^4 + 4x^2 + 4 - 4x^2 = x^4 + 4$.

D. Multiply the first terms and the last terms: $x^2 \cdot x^2 = x^4$ and $2 \cdot 2 = 4$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q4/5

Which argument proves the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$?

A. For n = 1, 2, and 3 the left side is 7, 19, and 37, and so is the right side, so it is true for every n.

B. $(n + 1)^3 = n^3 + 1$, so $(n + 1)^3 - n^3 = 1$, which is $3n^2 + 3n + 1$.

C. $(n + 1)^3 - n^3 = (n + 1 - n)^3 = 1^3 = 1$.

D. $(n + 1)^3 = n^3 + 3n^2 + 3n + 1$, so $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • MAIN • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 1

Expand the square of a binomial • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q1/5

Expand $(x^2 - y^2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q2/5

True or false?

$(a - b)^2 = a^2 - b^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q3/5

$(2xy)^2$ equals

A. $2xy^2$

B. $4xy$

C. $2x^2y^2$

D. $4x^2y^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q4/5

Expand $(n + 1)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 2

Recognize a difference of squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $a^2 - b^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q1/5

Factor $(x^2 + 2)^2 - (2x)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q2/5

True or false?

$u^2 - v^2 = (u - v)(u + v)$ for any expressions u and v.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q3/5

Which is a difference of squares?

A. $x^3 - 1$

B. $x^2 - 3x$

C. $9x^2 - 16$

D. $9x^2 + 16$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q4/5

Factor $25 - x^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSA-APR.C.4

BACKWARD 3

What makes an identity • Grade 6 • Expressions & Equations • 6.EE.A.4

Test $(a + b)^2 = a^2 + b^2$ with $a = 1$ and $b = 2$. Is it an identity?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q1/5

Explain the difference between an equation and an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q2/5

True or false?

Checking an identity with three numbers proves it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q3/5

$2(x + 3) = 2x + 6$ is

A. an identity

B. true only for $x = 3$

C. true only for $x = 0$

D. never true

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q4/5

Give values that show $a^3 + b^3 = (a + b)^3$ is not an identity.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 1

Prove or disprove proposed identities • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Is $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$ an identity? Which answer is correct?

A. Yes: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, and $3ab(a + b) = 3a^2b + 3ab^2$, so both sides are equal.

B. Yes, because $a = 1$ and $b = 1$ give $8 = 2 + 6$.

C. No: for $a = 1$ and $b = 2$ the left side is $3^3 = 27$ and the right side is $1 + 8 + 3 \cdot 2 = 15$, so the sides differ.

D. No: $(a + b)^3 = a^3 + b^3$, so the extra term makes the two sides different.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F1-Q1**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q1/5

Is $a^4 - b^4 = (a - b)^4$ an identity? Which answer is correct?

A. Yes: for $a = 1$ and $b = 0$ both sides equal 1.

B. No: for $a = 2$ and $b = 1$, the left side is $16 - 1 = 15$ and the right side is $1^4 = 1$.

C. Yes: a fourth power applies to each term inside the parentheses, so $(a - b)^4 = a^4 - b^4$ for all a and b.

D. No, because $a^4 - b^4 = (a - b)^2(a + b)^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q2/5

True or false?

$(x - 1)(x^2 + x + 1) = x^3 + 1$ is an identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q3/5

Which is an identity?

A. $(x + 2)^2 = x^2 + 4$

B. $(x + 1)^3 = x^3 + 1$

C. $x^2 - 4 = (x - 4)^2$

D. $(x + 2)^2 - (x - 2)^2 = 8x$

Find c so that $(x + c)^2 - (x - c)^2 = 12x$ is an identity.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSA-APR.C.4

FORWARD 2

Prove identities with the Binomial Theorem • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Which argument uses the Binomial Theorem to prove that the entries in row n of Pascal's Triangle add to 2^n?

A. Row n has n + 1 entries, and each entry is the sum of the two entries above it, so every row has twice as many entries as the row before it and the sum of row n is 2^n.

B. Row 3 is 1, 3, 3, 1, which adds to $8 = 2^3$, so every row n adds to 2^n.

C. Set $x = y = 1$ in $(x + y)^n = C(n, 0)x^n + C(n, 1)x^{n-1}y + \ldots + C(n, n)y^n$. The left side is 2^n and the right side is $C(n, 0) + C(n, 1) + \ldots + C(n, n)$, the row sum.

D. Set $x = 1$ and $y = 0$ in $(x + y)^n = C(n, 0)x^n + \ldots + C(n, n)y^n$: the left side is 1 and the right side is $C(n, 0) = 1$, so each row adds to 2^n.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S19-F2-Q1**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q1/5

Use the Binomial Theorem with $x = 1$ and $y = -1$ to find $C(5,0) - C(5,1) + C(5,2) - C(5,3) + C(5,4) - C(5,5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q2/5

True or false?

$(1 + 1)^4 = 1 + 4 + 6 + 4 + 1.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q3/5

The sum of the coefficients of $(2x + 1)^3$ is

A. 9

B. 8

C. 6

D. 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q4/5

Use $(1 + 0.1)^3 = 1 + 3(0.1) + 3(0.01) + 0.001$ to find 1.1^3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 19 • FORWARD 2 • Q5/5

Use polynomial identities to describe numerical relationships

MAIN	Algebra II main question	**HSA-APR.C.4**
BACKWARD 1	Substitute values into an identity	**6.EE.A.2**
BACKWARD 2	Check a Pythagorean triple	**8.G.B.6**
BACKWARD 3	Choose a and b to fit a product	**HSA-APR.C.4**
FORWARD 1	Generate a triple with a required side	**HSA-APR.C.4**
FORWARD 2	Connect the triple identity to complex numbers	**HSN-CN.A.2**

SET 20 • HSA-APR.C.4

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

The identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ generates Pythagorean triples. Use $x = 3$ and $y = 2$ to find a triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q1/5

Use the identity $(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$ with $x = 5$ and $y = 2$ to find a Pythagorean triple.

Triple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q2/5

Use the identity $(a + b)(a - b) = a^2 - b^2$ to compute 47×53 mentally. Show which a and b you use.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q3/5

Use the identity $(n + 1)^2 - n^2 = 2n + 1$ to write 41 as a difference of two consecutive squares.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q4/5

Use the identity $(n + 1)^3 - n^3 = 3n^2 + 3n + 1$ to compute $13^3 - 12^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • MAIN • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 1

Substitute values into an identity • Grade 6 • Expressions & Equations • 6.EE.A.2

Evaluate $x^2 - y^2$ for $x = 3$ and $y = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q1/5

Evaluate 2xy for x = 5 and y = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q2/5

True or false?

For $n = 12$, $3n^2 + 3n + 1 = 433$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q3/5

For $a = 50$ and $b = 3$, $a^2 - b^2$ is

A. 2491

B. 2500

C. 2494

D. 2509

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q4/5

Evaluate 2n + 1 for n = 20.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 2

Check a Pythagorean triple • Grade 8 • Geometry • 8.G.B.6

Do 5, 12, 13 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q1/5

Do 20, 21, 29 satisfy $a^2 + b^2 = c^2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q2/5

True or false?

6, 8, 11 is a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q3/5

Which is a Pythagorean triple?

A. 8, 15, 17

B. 6, 7, 9

C. 2, 3, 4

D. 4, 5, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q4/5

Explain why a triangle with sides 5, 12, 13 is a right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSA-APR.C.4

BACKWARD 3

Choose a and b to fit a product • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Write 47 and 53 as $50 - b$ and $50 + b$. What is b?

b = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q1/5

Write 98×102 as $(a - b)(a + b)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q2/5

True or false?

$29 \times 31 = 30^2 + 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q3/5

64 × 56 equals

A. $(60 - 4)^2$

B. $64^2 - 56^2$

C. $60^2 - 4^2$

D. $60^2 + 4^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S20-B3-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q4/5

Compute 98×102.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 1

Generate a triple with a required side • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4

Use the triple identity to find x and y (whole numbers, x > y) that give a triple with hypotenuse 25 and legs 7 and 24.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q1/5

Find x and y (whole numbers, x > y) for which the identity gives a triple with a leg 2xy = 40 and hypotenuse 41.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q2/5

True or false?

Every choice of whole numbers $x > y > 0$ in the identity gives a Pythagorean triple.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q3/5

Which choice gives the triple 8, 15, 17?

A. $x = 4, y = 1$

B. $x = 5, y = 3$

C. $x = 4, y = 2$

D. $x = 3, y = 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q4/5

Explain why $x = 2y$ gives a triple with legs in the ratio $3 : 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.4 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSA-APR.C.4

FORWARD 2

Connect the triple identity to complex numbers • High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2

Compute $(3 + 2i)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q1/5

Compute $(x + yi)^2$ in the form $a + bi$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q2/5

True or false?

The real and imaginary parts of $(x + yi)^2$ are the legs of the Pythagorean triple from the identity.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q3/5

$(5 + 2i)^2$ equals

A. 25 + 4i

B. 21 + 20i

C. 21 + 10i

D. 29 + 20i

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q4/5

Use $(4 + i)^2$ to find a Pythagorean triple.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Algebra II • Number & Quantity: The Complex Number System • HSN-CN.A.2 • Set 20 • FORWARD 2 • Q5/5

Expand with the Binomial Theorem

MAIN	Algebra II main question	**HSA-APR.C.5**
BACKWARD 1	Multiply binomials repeatedly	**HSA-APR.A.1**
BACKWARD 2	Build Pascal's Triangle by a rule	**4.OA.C.5**
BACKWARD 3	Powers of products and of negatives	**8.EE.A.1**
FORWARD 1	Find a specific term or coefficient with a condition	**HSA-APR.C.5**
FORWARD 2	Use binomial coefficients to compute probabilities	**HSS-CP.B.9**

SET 21 • HSA-APR.C.5

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Use the Binomial Theorem to expand $(x + 2)^5$. Write the result in standard form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q1/5

Use the Binomial Theorem to expand $(2a - b)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q2/5

Use the Binomial Theorem to find the coefficient of x^3 in $(x + 3)^7$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q3/5

Use the Binomial Theorem to find the term containing x^3y^3 in $(x - 2y)^6$.

Term: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q4/5

Use the first three terms of the binomial expansion of $(1 + 0.02)^5$ to estimate 1.02^5 to four decimal places.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • MAIN • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 1

Multiply binomials repeatedly • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Expand $(x + 2)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q1/5

Expand $(x + 2)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q2/5

True or false?

$(x + y)^3 = x^3 + y^3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q3/5

The number of terms in the expansion of $(x + y)^5$ is

A. 6

B. 5

C. 10

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q4/5

Expand $(a - b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 2

Build Pascal's Triangle by a rule • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

Row 4 of Pascal's Triangle is 1, 4, 6, 4, 1. Write row 5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q1/5

Write row 7 of Pascal's Triangle (row 6 is 1, 6, 15, 20, 15, 6, 1).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q2/5

True or false?

Each inner entry of Pascal's Triangle is the sum of the two entries above it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q3/5

The fourth entry of row 7 (counting from the first 1) is

A. 20

B. 7

C. 35

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q4/5

What is the sum of the entries in row 5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSA-APR.C.5

BACKWARD 3

Powers of products and of negatives • Grade 8 • Expressions & Equations • 8.EE.A.1

Simplify $(2a)^4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q1/5

Simplify $(-2y)^3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$(-b)^4 = -b^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q3/5

3^4 equals

A. 27

B. 12

C. 64

D. 81

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q4/5

Simplify $(0.02)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 1

Find a specific term or coefficient with a condition • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.C.5

Find the constant term of $(x + 1/x)^6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q1/5

The coefficient of x^2 in $(1 + kx)^5$ is 90. Find k ($k > 0$).

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q2/5

True or false?

The middle term of $(x + y)^8$ is $56x^4y^4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q3/5

The term with x^4 in $(x^2 - 1)^5$ is

A. $-5x^4$

B. $-10x^4$

C. $10x^4$

D. $5x^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q4/5

Find the coefficient of x in $(2x - 1)^6$.

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.C.5 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSA-APR.C.5

FORWARD 2

Use binomial coefficients to compute probabilities • High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.9

A fair coin is tossed 5 times. Use C(5, k) to find P(exactly 3 heads).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.B.9 • Set 21 • FORWARD 2 • Q1/5

A spinner lands on red with probability 0.3. Find P(exactly 2 reds in 4 spins).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.B.9 • Set 21 • FORWARD 2 • Q2/5

True or false?

The probabilities of 0, 1, ..., n successes add to 1 because $(p + (1 - p))^n = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.B.9 • Set 21 • FORWARD 2 • Q3/5

For 6 fair coin tosses, P(exactly 3 heads) is

A. 6/64

B. 3/6

C. 15/64

D. 20/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S21-F2-Q4**

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability
HSS-CP.B.9 • Set 21 • FORWARD 2 • Q4/5

Explain why C(n, k) counts the outcomes with exactly k successes in n trials.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School — Probability & Statistics (course placement in this bank) • Statistics & Probability: Conditional Probability & the Rules of Probability

HSS-CP.B.9 • Set 21 • FORWARD 2 • Q5/5

Rewrite rational expressions by polynomial division

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Write a division with a remainder as a mixed number	**4.NF.B.3**
BACKWARD 2	Multiply to check a division	**HSA-APR.A.1**
BACKWARD 3	Use the Remainder Theorem to check the remainder	**HSA-APR.B.2**
FORWARD 1	Divide with unknown coefficients and higher-degree divisors	**HSA-APR.D.6**
FORWARD 2	Use division to find slant asymptotes	**HSF-IF.C.7.d**

SET 22 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Which shows $(x^3 + 2x^2 - 5x + 3)/(x - 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^2 + x - 6 + 9/(x - 1)$

B. $x^2 + 3x - 2 + 1/(x + 1)$

C. $x^2 + 3x - 2 - 1/(x - 1)$

D. $x^2 + 3x - 2 + 1/(x - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q1/5

Which shows $(2x^3 - 3x^2 + 4x - 7)/(x^2 + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $2x - 3 + (2x + 4)/(x^2 + 1)$

B. $2x - 3 + (2x - 10)/(x^2 + 1)$

C. $2x - 3 + (2x - 4)/(x^2 + 1)$

D. $2x - 3 + (6x - 10)/(x^2 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q2/5

Which shows $(x^4 - 16)/(x - 2)$ written in the form $q(x) + r(x)/b(x)$?

A. $x^3 + 2x^2 + 4x + 8$ (remainder 0)

B. $x^3 + 2x^2 + 4x + 8 - 32/(x - 2)$

C. $x^3 - 2x^2 + 4x - 8$ (remainder 0)

D. $x^3 + 8$ (remainder 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q3/5

Which shows $(6x^2 + x - 4)/(2x + 1)$ written in the form $q(x) + r(x)/b(x)$?

A. $6x - 2 - 3/(2x + 1)$

B. $3x - 1 - 4/(2x + 1)$

C. $3x - 1 + 3/(2x + 1)$

D. $3x - 1 - 3/(2x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q4/5

Which shows $(x^3 - 4x + 5)/(x + 3)$ written in the form $q(x) + r(x)/b(x)$?

A. $x - 7 + 26/(x + 3)$

B. $x^2 - 3x + 5 - 10/(x + 3)$

C. $x^2 - 3x + 5 + 10/(x + 3)$

D. $x^2 + 3x + 5 + 20/(x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • MAIN • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 1

Write a division with a remainder as a mixed number • Grade 4 • Number & Operations–Fractions • 4.NF.B.3

Write 23/4 as a mixed number.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q1/5

Write 47/5 in the form q + r/5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q2/5

True or false?

17/3 = 5 + 1/3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q3/5